

Reconstruction-independent moment tests for quantum-computed dynamical spectra

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Abstract

We test spectra $A(\omega)$ reconstructed in quantum-sampled subspaces against ground-state moments computed without $A(\omega)$: a mismatch beyond a preset threshold rejects; a pass does not certify. In a self-attested, sealed blinded simulation (doped $L = 6$ Hubbard ring) the primary endpoint was 53.7% detection (101/188) at 0.9% false positives (1/112), low by construction; two sealed secondary predictions failed. At the deployed order (m_0-m_2) guaranteed power against fixed-pole redistributions is zero. On IBM Heron only the sampled support is device data; its missed weight Δ_0 rewards support spreading (a uniform in-sector sampler scores lower), and $\Delta_0 = 0.004$ coexists with relative L_1 error 0.35. Moments m_1, m_2 are simulation-only; a planned on-device test failed its noise gate.

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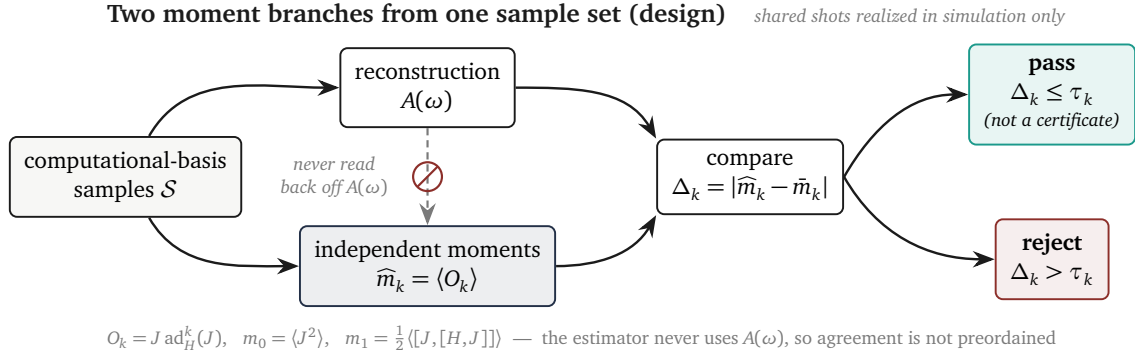


Figure 1: **Two moment branches from one sample set (design)**. Computational-basis samples fix a subspace $\mathcal{S}$. The reconstruction branch diagonalizes H in $\mathcal{S}$, assembles $A(\omega)$ and reads off $\bar{m}_k = \int \omega^k A(\omega) d\omega$. The estimator branch evaluates the ground-state operator moment $\hat{m}_k = \langle J(H - e_0)^k J \rangle$ with $e_0 = \langle H \rangle$ on the prepared state; on an exact eigenstate this equals $\langle J \text{ad}_H^k(J) \rangle$, the form printed in the drawing (Appendix A). It never uses $A(\omega)$, so it can disagree with the reconstruction. The reconstruction is rejected when $\Delta_k = |\hat{m}_k - \bar{m}_k|$ exceeds a threshold $\tau_k = z \delta_k$ fixed in advance from shot noise and a bias budget; a pass does not certify it. In this paper the two branches share one set of shots only in simulation; on hardware the device supplies the support $\mathcal{S}$ alone (Sec. 6).

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1 Introduction

Dynamical response functions such as $A(k, \omega)$, $S(q, \omega)$, $S^{zz}(q, \omega)$ and the optical conductivity, measured by photoemission, neutron and resonant x-ray scattering and infrared spectroscopy [1–4], are now computed on quantum processors first used for ground-state energies [5–9]: trapped ions have produced structure factors, spectral functions and Green’s functions built from measured Hamiltonian moments [10–12], and superconducting processors a DFT+DMFT impurity Green’s function with up to 14 qubits [13] and structure factors with up to 50 qubits, compared quantitatively with neutron data [14]. Sample-based diagonalization and quantum Krylov methods for ground states [15–19] have been extended to spectra [20, 21]: the device samples a subspace, and classical resources diagonalize the Hamiltonian in it and assemble $A(\omega)$ [22]. Our companion’s hardware run is exact by coverage and establishes only that this pipeline executes; its post-selection read the bitstrings in reversed bit

order, so all its retained determinants came from device errors [20] (Sec. 6).

Such results matter most where no trusted classical cross-check exists, a regime whose edge moves as classical methods improve and demonstrations are dequantized [23–28] (for sample-based diagonalization see Ref. [29]). That regime is the eventual target, not the setting of this paper: a check intended for it must first be calibrated where the ground truth is known [30–34], as we do here. Physics supplies the conditions: every $T = 0$ Lehmann spectrum obeys moment sum rules [35–37] and positivity [38], and the state behind it its conserved-charge identities [39], however it was computed.

Current practice validates a quantum-computed spectrum against a reference: neutron data [14], or the validation framework of Buchs *et al.*, with experimental and classical anchors and metrics that include low-order moments and sum-rule residuals; its asymmetric mode for the regime without full classical benchmarks, leaning more on experiment, sum rules and internal consistency, is left to future work [40]. Other verification tools, from self-verifying variational simulation to observable estimation without classical verification, certify a number, not a function, acting on states or expectation values [6, 41–53]. None checks a reconstructed $A(\omega)$ against the Hamiltonian that produced it without a classical or experimental reference. We target spectral weight at the wrong frequencies while integrated quantities stay within tolerance. Critical studies document the ground-state limitations of quantum-selected configuration interaction and subspace diagonalization [54–58]; that the same mechanisms misassign spectral weight is our extrapolation, and the same concern motivates response-targeted determinant selection on the classical side [59]. Figure 3 shows an injected corruption of $A(k, \omega)$ that passes visual inspection and a total-weight check but has the wrong first moment.

The screen compares two sets of moments (Fig. 1). For a zero-temperature Lehmann function of a probe J , the moments $m_k = \int \omega^k A(\omega) d\omega$ are ground-state expectations, $m_0 = \langle J^2 \rangle$, $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ and in general $m_k = \langle J(H - E_0)^k J \rangle$ [35, 38, 60]. Their estimate $\widehat{m}_k$ is never obtained by integrating $A(\omega)$, so it can disagree with the reconstruction’s moments $\bar{m}_k$; a moment read back off $A(\omega)$ gives $\Delta_k \equiv 0$. The reconstruction is rejected when $\Delta_k = |\widehat{m}_k - \bar{m}_k|$ exceeds a threshold τ_k fixed in advance from shot noise and a bias budget. These are necessary conditions: failing one rejects the reconstruction at the pre-registered level, while passing does not show that it is correct; it rules out only errors that move a tested moment past its threshold. Appendix D adds a state-level Gauss-law identity, evaluated exactly on the state vector [39, 41].

The design takes both branches from one set of samples; Sec. 2.1 states how far this is realized (not on hardware, where the device supplies only the sampled support; only in part, in simulation; and not in the blinded test).

Moment checks of computed spectra are long-standing in dynamical mean-field theory and quantum Monte Carlo, most systematically in the sum-rule program of Turkowski and Freericks [61, 62], and analytic-continuation pipelines report sum-rule agreement routinely [63, 64]; we claim no novelty for testing a spectrum against sum rules. The classical moment problem, with its orthogonal polynomials and Gauss quadrature [65–73], underlies the kernel-polynomial and recursion methods [74–77] and now bounds smeared spectral functions from Euclidean lattice correlators [78–80]; our moments come instead from ground-state operators, and our use is a rejection test on a real-frequency reconstruction. Semidefinite moment certificates [81, 82] would supply a second necessary condition on the reconstruction, not a sufficient one; we have not applied them (Sec. 9.1). Pipelines that hold the probe acting on the sampled ground state exactly, such as the Green’s-function solver of Patel *et al.* [21], never lose spectral weight; when both branches use the same ground-state vector, our coverage channel carries no signal for them, and only the higher moments test them.

This paper complements a companion study that reconstructs these spectra from sampled subspaces and bounds their error [20]; we adopt its terms (μ_j is its m_j). Its version 3 already

contains the captured weight w , giving our coverage residual $\Delta_0 = m_0(1 - w)$, the $k = 0$ operator leak of Eq. (B.3) with $|\varphi\rangle = J|0\rangle$; moment exactness through order $2K + 1$ for a subspace containing the order- K Krylov space, which is why Krylov reconstructions with two or more Lanczos nodes reproduce m_0 – m_2 exactly and pass, while the one-node case misses m_2 by 2–4% and passes only inside the 2% bias budget; the leakage identity $\mu_2 - \mu_2^S = \|Q_S H P_S \varphi\|^2$, the $k = 2$ leak on a subspace of unit captured weight; and the result that captured weight alone cannot certify a reconstruction, of which our $L = 8$ device run is an instance. We add not an error bound but a rejection test with shot-noise and bias-budget thresholds for any reconstructed single-particle, current or collective $A(\omega)$ (not only Rayleigh–Ritz ones), whose m_1 and m_2 detect weight moved in frequency at fixed total weight, which captured weight cannot see; a sealed, blinded calibration with its failures reported; the exact limits of what a pass implies, including zero guaranteed power at the deployed order; and a device evaluation of the coverage channel against reference distributions, showing that it cannot distinguish a good device from one that merely spreads its samples. In detail:

(i) *Screens across observables* (Figs. 3–6). In the doped Hubbard current response, m_1 (the mean frequency once normalized by m_0 [36, 37]) rejects a spurious low-frequency peak that preserves $m_0 = \langle J^2 \rangle$. The negative-order moment m_{-1} moves most in relative terms (154% against 38% for m_1) but, normalized by noise at a 2% bias, fires after m_1 and m_2 for this error model on the ring and the open chain, first only for scanned peaks at $0.5t$ and below; on the ring it also needs the Kohn stiffness from a separate calculation. The construction covers $A(k, \omega)$, $S(q, \omega)$ and $S^{zz}(q, \omega)$, all in exact simulation at $L \leq 12$.

(ii) *Independence under truncation* (Sec. 4). Under determinant-subspace truncation the independent estimator crosses its threshold while a control read back off $A(\omega)$ stays identically zero [15, 16]. The same holds when both branches are evaluated on one subspace state.

(iii) *A blinded, pre-registered calibration* (Sec. 5). Before any instance existed we sealed the battery, thresholds, error catalog and seed by a SHA-256 hash; the seal is self-attested, with no third-party timestamp. The sealed primary endpoint was 53.7% detection (101/188, Wilson 95% interval [46.6, 60.7]%) at 0.9% false positives (1/112, [0.2, 4.9]%) : 56/56 determinant truncations (all 56 also rejected in a post hoc shared-state rescoring), 45/61 spurious features and 0/71 under-converged Krylov reconstructions were caught. Sealed secondary predictions held for weight-changing features (25/27) and multi-node Krylov (0/56); two failed: one-node Krylov, expected to be caught, was caught in 0/15, and weight-preserving features, expected to be caught rarely, in 20/34 (59%), which was not anticipated. The false-positive rate is low by construction: the simulated estimator is unbiased while the threshold budgets a 2% bias, and a realized +2% (–2%) bias would raise the modeled rate from 1.3% to 8.0% (11.5%); the 0.9% confirms the implementation, not robustness to a real mitigation bias, which is untested. The catalog is one doped $L = 6$ ring ($U/t \in \{3, 4, 5, 6\}$, current probe) whose random truncations keep 75–99% of the ground-state weight; the 56/56 describes this sealed draw, not every truncation: an exhaustive $U/t = 4$ scan finds 12 of 185 depths that pass the full battery, three (46–51% of the support) inside the catalog’s depth range.

(iv) *Exact limits* (Secs. 7 and 8). Classical Christoffel–Markov–Stieltjes and Markov–Krein brackets, re-derived here [69, 83], bound the cumulative weight of a reconstruction with exact moments; at the deployed order they are vacuous at the mean ($\max_t W_1 = m_0$), and finite tolerances widen them. The first N moments are complete for fixed-pole weight redistributions on N distinct poles, in exact arithmetic only (Theorem 2); at the deployed order and $N > 3$ the guaranteed power is zero, because the redistributions preserving m_0, m_1, m_2 form an $(N - 3)$ -dimensional space, whatever the pole separation. On the $L = 6$ current spectrum ($N = 5$) redistributions that pass move 72% of the weight (keeping m_0 – m_2 exact) and 86% (shifting m_2 by exactly τ_2). Closing the gap needs moments beyond m_2 , at a shot cost growing at least as $D^{-2(s-1)}$ for weight moved within a cluster of s poles of diameter D .

Table 1: **Contribution relative to the closest prior art.** “recon.-indep.”: compared with something the reconstruction did not produce, and with what; “same samples”: comparison quantity taken from the spectrum’s own samples; “—”: the axis does not apply. The semidefinite programs bound the true moments; read-off moments inside their intervals are a second necessary condition, not a sufficient one (Sec. 9.1).^aDesigned; realized only in part, and only in simulation (Sec. 2.1); same-sample estimation, blinded pre-registration and on-device execution were not realized together.^bSampled support only; m_1, m_2 are simulation-only (Table 2). ^cBlinded simulation with an oracle-centred estimator (Sec. 5). ^dThe companion’s $L = 6$ run covers the whole sector, so its bound is not exercised on device data.

method	recon.-indep.	same samples	on device	blinded	role
Turkowski–Freericks sum-rule checks [61,62]	yes (vs classical)	no	no	no	consistency check
maxent / stochastic continuation [63,64]	yes (vs classical)	no	no	no	consistency check
Buchs <i>et al.</i> validation framework [40]	yes (vs experiment, classical, sum-rule targets)	no	no	no	validation metrics, incl. moments
Lee <i>et al.</i> neutron benchmark [14]	yes (vs experiment)	no	yes	no	benchmark
Greene-Diniz <i>et al.</i> cumulant Lanczos [12]	— (moments build G)	—	yes	no	constructive
Wang–Acín / Mortimer SDP [81,82]	— (bounds moments)	no	no	no	two-sided moment bounds
Abbott <i>et al.</i> / causal bootstrap [78–80]	yes (Euclidean)	no	no	no	two-sided bound
Kokail <i>et al.</i> self-verification [41]	— (certifies state)	—	yes	no	state verification
reference-free observable estimation [52]	— (certifies number)	—	yes	no	validity
companion: captured weight and leakage [20]	— (bounds the error)	—	no ^d	no	a posteriori error bound
this work	yes	designed^a	support only^b	yes^c	necessary condition

(v) A device evaluation of the coverage channel (Sec. 6). On IBM Heron (ibm_fez) we ran seven circuits for an $L = 8$ electron-addition sector at four budgets (5×10^4 to 4×10^3 shots per circuit) and kept the raw counts; the circuits sample the addition sector but are not the Hubbard propagator of the probe state. The device supplies only the support $\mathcal{S}$. The missed weight Δ_0 grows from 0.0044 to 0.37 as coverage falls from 94% to 35% but measures support, not fidelity: the device lies below all noiseless replicas of its circuits at 5×10^4 shots and above all of them at 4×10^3 , and a uniform in-sector sampler at the device’s retained count reaches a lower Δ_0 and a more accurate reconstruction at every budget. The reconstruction with $\Delta_0 = 0.0044$ has a relative L_1 error of 0.35. The moments m_1, m_2 were not measured on the device: at the companion’s depth (292 CZ) the forecast depolarizing bias of the bond-basis m_1 estimate is 37–116% of the signal (local to global, two-qubit error 2.5×10^{-3}). A sealed simulated noise gate for an on-device momentum-distribution test failed at the relaxed floors of its single permitted amendment ($T_1 = 100 \mu\text{s}$, $T_2 = 70 \mu\text{s}$, two-qubit error parameter 3.5×10^{-3}), so that run was not made.

Scope. The method can reject a reconstruction but cannot certify one; its conditions are necessary, not sufficient. The moment screen targets determinant-subspace truncation and

spurious features in SQD/QSCI spectra (weight and subspace misassignment); the Gauss-law identity targets gauge-symmetry violation in the prepared state. Krylov-convergence errors are out of scope: by the moment exactness stated above, an under-converged Krylov reconstruction passes the battery (Sec. 5), and detecting that error class calls for a separate instrument, most naturally a Ritz-residual check on the Krylov basis, which we leave to future work. Neither check is a general device-noise diagnostic [52]. The screen tests integrated moments of diagonal $T = 0$ Lehmann spectra [38], not pointwise line shapes; every demonstration is classically reproducible ($L \leq 12$ Hubbard chains and rings, a small lattice-Schwinger model, a DMRG check at $L = 24$); and the blinded test withholds a classically known truth, so it is not a field catch, and no device is in the loop. On hardware, where moments carry statistical and systematic error, interval tests must replace the exact-moment tests (Appendix B); Table 2 separates what is measured on hardware from what is simulated or deferred.

2 Method: reconstruction-independent moment tests

We first describe the object the screen tests, then the checks attached to it.

2.1 The quantum processor and the origin of the samples

The screen tests the output of a sample-based-diagonalization pipeline. The pipeline and the $L = 6$ device spectrum of Fig. 11 are the companion study's [20]; this paper contributes the test, its blinded calibration in simulation, its exact limits and one $L = 8$ ibm_fez job that supplies a device-sampled support (Sec. 6). In sample-based diagonalization [15–17] the processor does not evaluate $A(\omega)$: each computational-basis shot of a prepared state returns a bitstring (a candidate Slater determinant), and the distinct bitstrings surviving post-selection to the physical particle-number and spin sector form the subspace $\mathcal{S}$. The Hamiltonian is projected onto $\mathcal{S}$ and diagonalized classically, and the spectrum is the $T = 0$ Lehmann sum $A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta_\eta(\omega - (E_n - E_0))$ over the resulting eigenpairs, with δ_η a narrow broadening [38]. In the companion's formulation the prepared state is a Trotterized real-time evolution of the $(N+1)$ -particle seed $c_p^\dagger|\psi_0\rangle$ [84]. The device circuits actually run, the companion's $L = 6$ job and our $L = 8$ job, are number- and S_z -conserving samplers of the addition sector, not the Hubbard propagator of the seed, as the companion's v3 discloses for its run (Sec. 6) [20]. Neither run used configuration recovery [29], and the companion's $L = 6$ run read the bitstrings in reversed qubit order during post-selection, so every determinant it retained arose from a device error [20].

The intended role of the processor appears only at scale, where the configurations carrying the weight of the probe state $|\varphi\rangle = J|0\rangle$ cannot in general be enumerated classically: an ideal sampler of the targeted state would concentrate shots on them and fix $\mathcal{S}$ by measurement rather than by a classical selection heuristic [16]. We do not show that any device sampler does this better than a classical heuristic, and the companion claims no quantum advantage [20]. The circuits run here are not targeted at φ : for them the coverage residual Δ_0 (Sec. 6) measures the weight of φ outside the configurations they happened to reach, and at $L = 8$ a uniformly random in-sector sampler, given the device's retained shot counts, leaves less of that weight uncovered than the device at every shot budget.

The design goal is that the moment estimator and the reconstruction share the sampled support, so that $\widehat{m}_k$ inherits the shot noise and mitigation choices of the spectrum it tests. The estimator $\widehat{m}_k$ defined below applies the full operator to the subspace ground state built on $\mathcal{S}$, a classical post-processing step that reaches configurations outside $\mathcal{S}$; a device-native alternative, measuring $\langle O_k \rangle$ in qubit-wise-commuting Pauli groups on a separately prepared ground state, is simulated only (Appendix B). We state where the shared-sample design is

209 realized. On hardware it is not: the device supplies only the support $\mathcal{S}$, and every moment in
 210 Sec. 6 is computed classically. In simulation it is realized in three places: the shared-state test
 211 of Fig. 8, where both moments are evaluated on one subspace state (a deterministic truncation
 212 by $|\psi_0|^2$, not a set of shots); the joint-sampling simulations of Sec. 9 and Appendix B, where
 213 one set of shots yields both $\widehat{m}_1$ and $\bar{m}_1$ (correlation $\rho \approx 0.43$ at 84% coverage, no gain in
 214 detection power; the local estimator uses exact amplitudes and is not device-realizable); and
 215 the $L = 24$ DMRG test of Sec. 9, where both branches use one sampled subspace. The blinded
 216 test of Sec. 5 does not use it: there $\widehat{m}_k$ is the exact moment plus Gaussian shot noise; that
 217 section also reports a post hoc rescoring of its 56 truncations with the shared-state estimator.

218 2.2 The independent moment estimator and the falsifier

219 For a zero-temperature diagonal Lehmann spectral function of an operator J , $A(\omega) =$
 220 $\sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0))$ is a positive measure on the real line. On an exact
 221 eigenstate its power moments $m_k = \int \omega^k A(\omega) d\omega$ are ground-state expectation values,
 222 $m_k = \langle 0|J(H - E_0)^k J|0\rangle = \langle 0|J \text{ad}_H^k(J)|0\rangle$, so that $m_0 = \langle J^2 \rangle$ and $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$
 223 (Appendix A) [35, 38]. In the optical case m_0 is the total current-current weight and m_1/m_0
 224 the mean frequency; the optical f -sum is the separate moment m_{-1} [36, 37]. These operator
 225 forms coincide only on an exact eigenstate; on a subspace state they differ by enough to
 226 change a verdict (Appendix A). We therefore fix the deployed estimator as $\widehat{m}_k = \langle J^\dagger (H - e_0)^k J \rangle$
 227 with $e_0 = \langle H \rangle$ on the prepared state, the form used in the shared-state tests below.

228 The screen attaches to the spectrum an estimator $\widehat{m}_k$ of each moment that is independent
 229 of the reconstruction: it is never obtained by integrating $\omega^k A(\omega)$. The reconstruction moment
 230 $\bar{m}_k = \int \omega^k A(\omega) d\omega$ is the quantity a circular check would use, and the test statistic is the
 231 discrepancy $\Delta_k = |\widehat{m}_k - \bar{m}_k|$. The true spectrum gives $\Delta_k = 0$ up to estimator error, so the
 232 reconstruction is rejected when $\Delta_k > \tau_k$, with $\tau_k = z \delta_k$, $z = 1.96$, and $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$
 233 the shot variance plus a bias budget b_k (Appendix B); the deployed rule uses $N_s = 5 \times 10^4$ and
 234 $b_k = 0.02 |\widehat{m}_k|$. A circular check compares $\bar{m}_k$ with itself, returns $\Delta_k \equiv 0$ and can never reject;
 235 Results II shows this contrast on controlled truncation. At $k = 0$ the discrepancy is the weight
 236 the support misses, $\Delta_0 = \langle \varphi | (\mathbb{1} - \Pi_{\mathcal{S}}) | \varphi \rangle = m_0(1 - w)$, the companion's captured-weight
 237 deficit [20]; the companion's moment-exactness theorem and m_2 leakage identity [Eq. (B.3)
 238 at $k = 2$] bound what the test can see, and what this paper adds is stated in Sec. 1.

239 The screen is a battery, not a single test. It applies Δ_k for $k = 0, 1, \dots, K$ (deployed: $K = 2$)
 240 and, separately, checks that $\{\widehat{m}_k\}$ is itself a moment sequence of a positive measure: the Hankel
 241 matrix $H_n = [\widehat{m}_{i+j}] \succeq 0$ and, because a diagonal $T = 0$ Lehmann measure is supported on
 242 $\omega = E_n - E_0 \geq 0$, also the shifted Stieltjes matrix $H_n^{(1)} = [\widehat{m}_{i+j+1}] \succeq 0$ [65–67] (Fig. 2).
 243 A violation rejects the estimate—device, mitigation or estimator—independently of $A(\omega)$. It
 244 cannot reject a non-negative reconstruction, whose own moments satisfy these conditions by
 245 construction; in the blinded test the branch was applied to $\bar{m}_k$ and could not fire (Sec. 5). The
 246 battery also accepts any exactly known operator identity, such as a conserved charge $[H, G] = 0$
 247 with a known physical eigenvalue, whose violation rejects the prepared state (gauge-theory
 248 example below) [39]. The construction transports across models and observables because it
 249 needs only (a) a positive spectral measure and (b) operators whose ground-state expectations
 250 can be estimated without the reconstruction [69, 74].

251 3 Results I: weight-misassignment errors across observables

252 Figure 3 shows $A(k, \omega)$ of the half-filled Hubbard ring, a one-dimensional correlated insulator
 253 with Hubbard bands separated by the Mott gap [3, 85–88]. The single-particle gap approaches

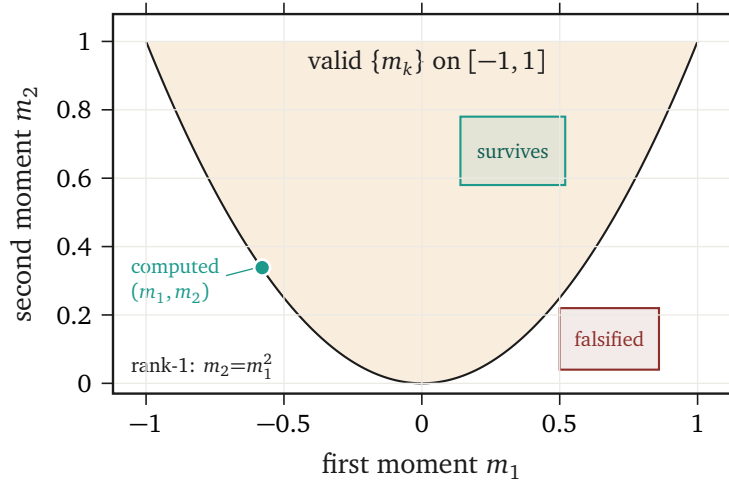


Figure 2: **Hankel-positivity region in the (m_1, m_2) plane.** With $m_0 = 1$ and frequencies rescaled to $[-1, 1]$ as in Fig. 13 (Appendix C), moments lie in the body $m_1^2 \leq m_2 \leq 1$, bounded below by the Hankel (rank-one) edge of a single δ and above by the interval (Hausdorff) localizing condition. Of two schematic shot-budget boxes, the oxblood one (“falsified”), wholly Hankel-negative, rejects the estimate without reference to $A(\omega)$; the teal one (“survives”) passes. The check acts on the estimate: a non-negative reconstruction’s own moments always lie inside the body. Teal dot: the doped $L = 6$, $U/t = 8$ current measure (exact moments); one pole carries 78% of its weight, so the dot lies just above the rank-one edge [65, 66].

254 U only in the atomic limit $t \rightarrow 0$, and at the size ($L = 12$) and broadening of the figure the
 255 spinon and holon branches are only partly resolved. An injected corruption flattens the band-
 256 centroid dispersion by moving weight between the Hubbard bands at each momentum, leaving
 257 the Mott gap and the per- k total weight intact. For this probe m_0 is pinned to 1, so the first
 258 informative moment is m_1 [38, 62]: from the anticommutator sum rule for the non-Hermitian
 259 probe $c_{k\sigma}$, measured from $U/2$, it is the Hartree centroid $\epsilon_k + U(\langle n_{-\sigma} \rangle - \frac{1}{2}) = \epsilon_k$ at half filling
 260 (Appendix A), and it exposes the flattening (right panel). The corruption moves the centroid
 261 by a fixed fraction of ϵ_k and the moment is closed-form, so the residual is set by construction:
 262 the panel illustrates the check rather than testing it.

263 Figure 4 shows the collective response: the charge structure factor $S(q, \omega)$, gapped by a
 264 particle-hole onset Δ_c , and the spin structure factor $S^{zz}(q, \omega)$, whose gapless spinon contin-
 265 uum [89] signals spin-charge separation [3]. The onset Δ_c is a finite-size quantity distinct
 266 from the single-particle Mott gap ($\approx 4.97 t$); both are set by U and coincide only in the ther-
 267 modynamic limit. The charge (spin) data, on the scale $\sim U$ ($\sim J$), are computed to $11 t$
 268 ($2.6 t$), displayed to $10 t$ ($2.4 t$) and linearly interpolated in q . Each channel is a positive
 269 $T = 0$ Lehmann measure with its own moment sum rules [35], so the screen applies to each
 270 (Fig. 6). Such spectra are standard output of classical many-body methods [63, 90–93]; at
 271 these sizes we compute them exactly.

272 We now apply the screen to a wrong distribution of correct total weight. In the doped
 273 one-dimensional Hubbard model the regular ($\omega > 0$) current spectral function $A_J(\omega)$ is dom-
 274 inated by a mid-infrared/Mott band of total weight $m_0 = \langle J^2 \rangle$ [2, 36, 37, 96]; on the finite ring
 275 the charge stiffness is a separate zero-frequency Drude term, absent from $A_J(\omega)$. A wrong
 276 reconstruction moves a fraction f of the weight above $6 t$ into a spurious Drude-like pole
 277 at $2.5 t$, keeping the total fixed, so a total-weight check passes it (Fig. 5; Lanczos ground
 278 state, Haydock/Ritz current poles [73, 75], every moment exact, no sampling). The first mo-
 279 ment $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ [35, 36] differs from the reconstruction’s by up to 38% of m_1 at

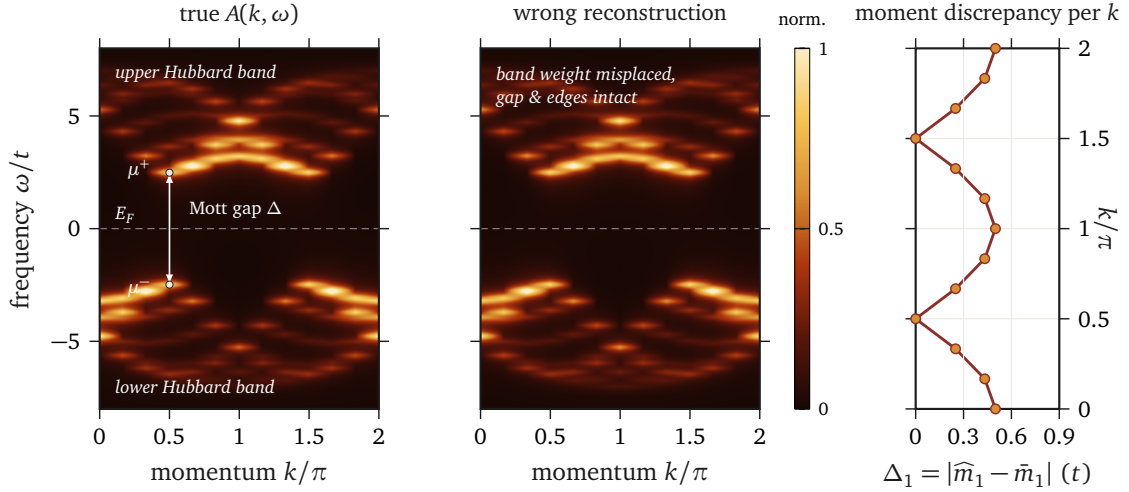


Figure 3: **The first-moment check on the single-particle spectral function $A(k, \omega)$ of the Hubbard model.** Left: exact $A(k, \omega)$ of the half-filled ring, frequencies measured from $E_F = U/2$ (dashed); the band edges $\mu^\pm = \pm 2.484 t$ at $k_F = \pi/2$ (dots, arrow) give the Mott gap $\Delta = \mu^+ - \mu^- = 4.97 t$. Center: an injected corruption rebalances intensity between the Hubbard bands at each momentum so that the band centroid becomes $(1 - \alpha)\epsilon_k$, with $\alpha = 0.25$. The gap stays free of poles (only the Lorentzian tail appears at E_F), $\mu^\pm$ are unchanged and $m_0 = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$ per k , so a total-weight check accepts it. Right: $\Delta_1 = |\widehat{m}_1 - \bar{m}_1|$, with $\widehat{m}_1(k) = \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle$ the closed-form Hartree centroid, not a sampled estimate; $\Delta_1 = \alpha|\epsilon_k|$ by construction ($0.50 t$ at $k = 0, \pi, 2\pi$; zero at $k = \pi/2, 3\pi/2$). $L = 12$ (12 momenta), $U/t = 8$, $\eta = 0.16 t$; sector-basis Lanczos with Haydock Green's functions [73, 75].

$f = 0.5$ and, under the deployed threshold rule (2% bias budget at $N_s = 5 \times 10^4$), rejects from $f \approx 0.057$. Integrated weight is a weak check for this error.

The misplaced weight also moves $m_{-1} = \int_0^\infty A_J(\omega)/\omega d\omega = \sum_n |\langle n|J|0 \rangle|^2/\omega_n$, whose kernel $1/\omega$ weights low frequencies, unlike the kernel ω^k of the positive moments (Fig. 5). Since $J = i[H, P]$, this optical f -sum moment is a ground-state kinetic energy: $m_{-1} = \frac{1}{2}\langle -\hat{T} \rangle$ on open chains and $\frac{1}{2}\langle -\hat{T} \rangle - \mathcal{D}$ on the ring, with $\mathcal{D}$ the finite-size Drude weight (derived and verified in Appendix A) [35–37]. On this doped ring only 1.4% of m_0 lies below $6 t$, and $m_{-1} = 0.324$ is the small difference of $\frac{1}{2}\langle -\hat{T} \rangle = 5.025$ and $\mathcal{D} = 4.701$: the stiffness cancels 94% of the kinetic term. The spurious peak therefore moves m_{-1} by up to 154% at $f = 0.5$, $4.0\times$ the shift of m_1 (38%) and $3.3\times$ that of m_2 (47%).

A larger relative shift does not mean earlier detection. Because of the cancellation, a 2% bias on $\frac{1}{2}\langle -\hat{T} \rangle$ sets $\tau_{-1} = 0.198$, which is 61% of m_{-1} , whereas τ_1 and τ_2 are 4.3% of m_1 and m_2 . So m_2 rejects first ($f \approx 0.046$), then m_1 (0.057), and m_{-1} only at $f \approx 0.20$. The open chain gives the same order, and across spurious-pole positions of $0.25\text{--}4 t$ the moment m_{-1} fires first only for poles at $0.25 t$ and $0.5 t$, on both geometries (Appendix A). It is thus a complementary check for weight very close to $\omega = 0$, not the most sensitive check for this error model. On the ring, $\mathcal{D}$ requires ground states at twisted flux and is not a same-sample quantity. We claim no novelty for the identity, the textbook f -sum [36, 37].

On the collective channels (Fig. 6), for each probe $A \in \{\rho_q, S_q^z\}$, with $\rho_q = \sum_j e^{-iqj} n_j$ and $S_q^z = \sum_j e^{-iqj} S_j^z$, both moments are ground-state expectations, $m_0(q) = \langle A^\dagger A \rangle_c$ and $m_1(q) = \langle 0|A^\dagger(H - E_0)A|0 \rangle = \frac{1}{2}\langle [A^\dagger, [H, A]] \rangle$. On a half-filled $L = 6$ ring at $U/t = 8$ the operator moments reproduce the exact Lehmann moments of $S(q, \omega)$ and $S^{zz}(q, \omega)$ to better than 10^{-14}

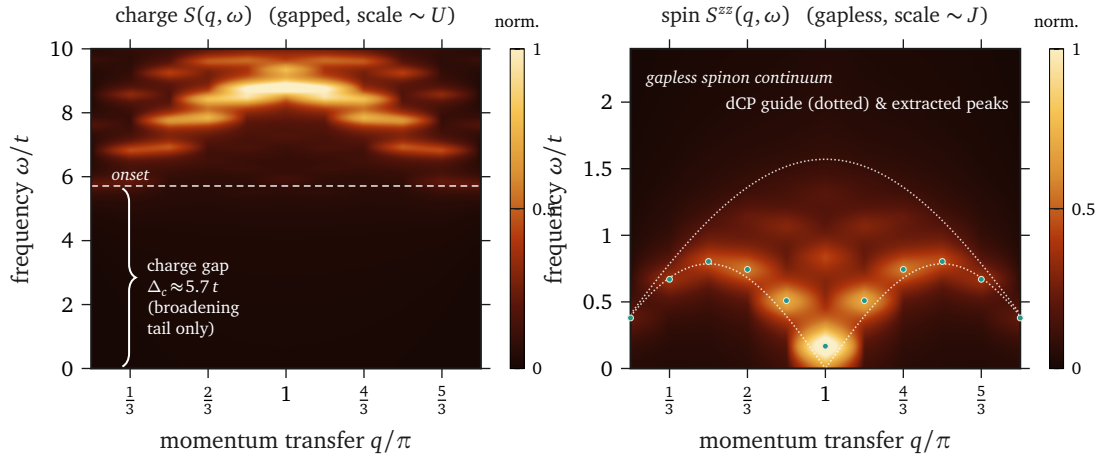


Figure 4: Charge and spin structure factors of the half-filled Hubbard ring. Left: the charge structure factor $S(q, \omega)$ is gapped by a particle–hole onset $\Delta_c \approx 5.7 t$ (dashed: lowest frequency above 5% of the peak intensity); below it only the broadening tail appears. Right: the spin structure factor $S^{zz}(q, \omega)$ carries gapless spinon weight. White dotted: des Cloizeaux–Pearson (dCP) two-spinon boundaries $\omega_L = (\pi J/2)|\sin q|$ [94] and $\omega_U = \pi J|\sin(q/2)|$ [95], $J = 4t^2/U = 0.5 t$, an ideal large- U guide, not a strict bound (finite U , size and broadening put some weight below ω_L). Teal points: dominant spinon peak at the eleven momenta $q/\pi \in \{1/6, \dots, 11/6\}$, close to the lower edge. No moment test is applied here; Fig. 6 applies it at $L = 6$. $L = 12$, $U/t = 8$, $\eta = 0.20 t$ (charge), $0.12 t$ (spin); sector-basis Lanczos, Haydock evaluation [73, 75].

at every momentum, and the two forms of m_1 , equal on the exact ground state (Appendix A), agree to the same precision. A reconstruction that moves band weight into a spurious low-frequency peak at fixed total weight passes the m_0 check (residual $\sim 10^{-15}$), but its first moment is off by 29–36% once 40% of the band weight is moved, for both observables at $q/\pi = 1/3, 2/3, 1$. The mean excitation frequency m_1/m_0 registers the two energy scales, set by U for charge ($\approx 8.5 t$ at $q = 2\pi/3$) and by $J = 4t^2/U$ for spin ($\approx 1.0 t$). These are exact simulations at $L = 6$, not the $L = 12$ spectra of Fig. 4, and not a hardware result.

The battery also admits exactly known operator identities that screen a different error class. The $U(1)$ lattice-Schwinger Gauss law $\langle \sum_x G_x^2 \rangle = 0$ detects gauge-symmetry breaking; its operator is diagonal in the computational basis and so estimable from samples in principle, but here it is evaluated exactly on the state vector, and it checks the state, not a reconstruction (Appendix D) [39, 41].

4 Results II: independence supplies the discriminating power

The method’s central claim is that *independence* gives the screen its discriminating power. We test it on the native failure mode of sample-based diagonalization, determinant-subspace truncation, in which the sampled subspace omits configurations that carry spectral weight and the reconstruction is internally consistent with an incomplete basis [15, 16]. Two checks run on the same truncated data (Fig. 7). The independent estimator computes $\widehat{m}_1 = \langle O_1 \rangle$ with the full operators J and H , not their restriction to the retained determinants (here on the exact full-sector ground state; the idealization is removed below). The back-computed control $\bar{m}_1$ integrates the reconstructed $A(\omega)$, which lives inside the truncated subspace; this truncated

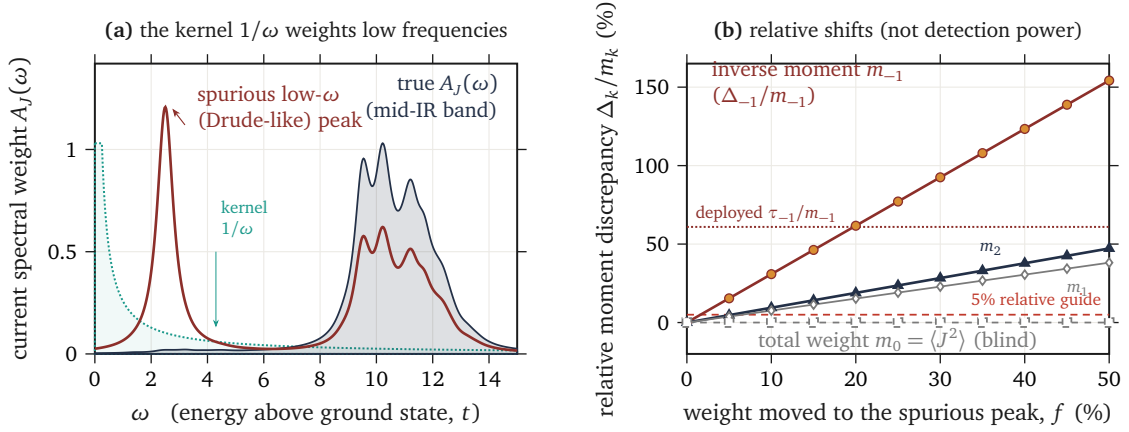


Figure 5: **Current response: misplaced weight at fixed total.** Doped Hubbard ring ($L = 12$, $U/t = 8$, filling $2/3$; $\text{Re } \sigma \propto A_J/\omega$). A weight fraction f above $6t$ moves to a spurious pole at $2.5t$ at fixed $m_0 = \langle J^2 \rangle$. (a) True spectrum (indigo; 1.4% of m_0 below $6t$) and spurious peak (oxblood), where the kernel $1/\omega$ (teal) is large. (b) Exact Δ_k/m_k versus f : $\Delta_0/m_0 < 10^{-13}$; m_1 , m_2 and $m_{-1} = \frac{1}{2}\langle -\hat{T} \rangle - \mathcal{D} = 0.324$ move by up to 38%, 47% and 154%. The 5% line is a common reference, not the deployed threshold. Deployed thresholds: 4.3% of m_1 and m_2 , but 61% of m_{-1} (dotted), because of a 94% cancellation of $\frac{1}{2}\langle -\hat{T} \rangle$ by the Kohn stiffness $\mathcal{D}$; rejection at $f \approx 0.046$ (m_2), 0.057 (m_1), 0.20 (m_{-1}). $\mathcal{D}$ is from the flux curvature, not the same samples.

first moment is $\langle J\psi_0^d | (H_d - E_0^d) | J\psi_0^d \rangle$, obtained from one subspace ground state without full diagonalization [73]. As truncation deepens, Δ_1 grows and crosses an illustrative fixed 5% threshold, whereas the circular control compares $\bar{m}_1$ with itself and is zero by definition [16]. Hence every moment in the screen must be estimated independently of $A(\omega)$, never read back off it [15].

Because the independent estimator applies the exact O_k to the prepared state, Δ_k detects weight that leaks *outside* the retained determinant support, not a redistribution among the retained poles that preserves the tested moments. In exact arithmetic this blind spot closes at order $N - 1$ on N distinct poles (Theorem 2, Appendix E). At the deployed order, however, the guaranteed power against such redistributions is zero, by dimension count rather than conditioning: every redistribution in the $(N - K - 1)$ -dimensional null space of the moment rows $0, \dots, K$ passes, whatever its size. For the $U/t = 4$ current measure of the blinded test (Sec. 5; $N = 5$ weighted poles, $K = 2$, a two-dimensional null space) one such redistribution moves 72% of the weight, empties two poles and passes, also staying below the m_3 and m_4 thresholds at the deployed budget (Sec. 8). The screen is thus a necessary-condition detector of support leakage and of weight misassignment that shifts the low moments; redistributions that preserve the deployed moments are a disclosed blind spot, not an error class we claim to catch.

Figure 7 is an idealization: a sample-based pipeline has only the subspace state built from the sampled support, and both branches must be computed from it. We therefore evaluate both moments on the *same* ground state $|0_S\rangle$ of $H_S = \Pi_S H \Pi_S$, with shared reference $e_0 = \langle 0_S | H | 0_S \rangle$ (Fig. 8). The independent estimator $\hat{m}_1 = \langle 0_S | J(H - e_0) J | 0_S \rangle$ applies the *full* sparse operators, reaching determinants outside $\mathcal{S}$; the reconstruction moment $\bar{m}_1 = \langle 0_S | J_S (H_S - e_0) J_S | 0_S \rangle = \int \omega A(\omega) d\omega$ stays within $\mathcal{S}$. This form, $\langle J(H - e_0)^k J \rangle$ with e_0 the energy of the prepared state, is the estimator we fix throughout; off an eigenstate it differs from the double-commutator form (Appendix A). The shared-state estimator needs only the sampled

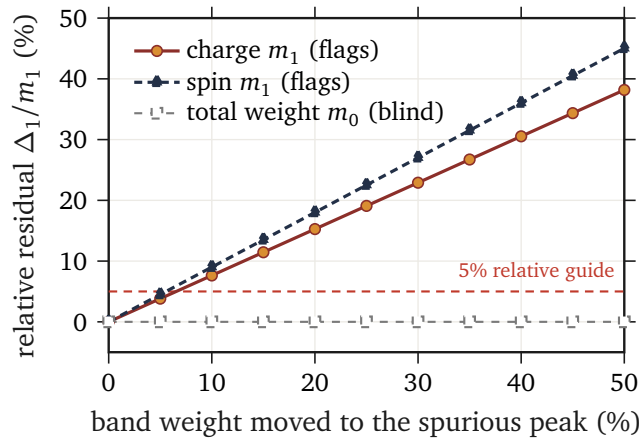


Figure 6: **Moment test on the charge and spin channels.** The screen of Fig. 5 applied to $S(q, \omega)$ and $S^{zz}(q, \omega)$ of a half-filled Hubbard ring ($L = 6$, $U/t = 8$; exact simulation). A wrong reconstruction moves band weight into a spurious peak at $\omega = 2.0t$ (charge gap) or $0.10t$ (spin), conserving m_0 exactly. At $q/\pi = 2/3$, m_0 (grey) stays blind for both channels, while the relative first-moment residual of the charge (ρ_q , oxblood) and spin (S_q^z , indigo) probes grows linearly and crosses a 5% guide, not the deployed τ_1 . Operator and exact Lehmann moments agree to better than 10^{-14} at every momentum $q/\pi \in \{1/3, 2/3, 1\}$. $L = 12$ spectra: Fig. 4.

support and is evaluated classically; it is not a moment measured on a device. The difference $\hat{m}_1 - \bar{m}_1$ is the operator leak of Eq. (B.3) at $k = 1$; its $k = 0$ and $k = 2$ forms are the companion's captured-weight and boundary-coupling results, stated there for the probe on the exact ground state [20], and its use as a thresholded test against the circular control is this paper's (Sec. 1). The shared-state discrepancy rises by about five orders of magnitude, from 5×10^{-6} at full support to $\approx 100\%$ at deep truncation (non-monotone near both ends: below 10^{-4} near full support, and at saturation). It crosses the 5% line at $d \approx 4.4 \times 10^4$ (18% coverage); the idealized curve crosses at $d \approx 1.6 \times 10^4$ (7%) after a dip at moderate truncation, where the full-state reference and the truncated reconstruction happen to track each other. With the shared state the screen thus fires at milder truncation: at this threshold the idealization does not inflate the result. Neither branch stays exact under truncation: $\hat{m}_1$ rises from 35.1 at full support to 60.1 at the deepest cut, while $\bar{m}_1$ falls toward zero. The discrepancy measures the leak, not the error of either branch against the exact moment.

5 A pre-registered test protocol

To probe the screen on errors unknown to the classifier, we ran a blinded, pre-registered simulation. Before any instance existed we sealed by SHA-256 hash (0e4c3368...) the battery (m_0, m_1, m_2 plus a Hankel-positivity condition), the decision rule and thresholds, the error catalog, the primary endpoint, the secondary predictions and the random seed. The rule is that of Sec. 2.2 (reject when any $|\hat{m}_k - \bar{m}_k| > \tau_k$, $k \leq 2$, or the Hankel condition fails; $z = 1.96$; $N_s = 5 \times 10^4$, the largest per-circuit budget of the `ibm_fez` runs; assumed bias $b = 2\%$; $\text{Var}(O_k)$ the local-estimator variance, Appendix B). Generator and classifier re-verify the hash and refuse to run if it is broken. The seal is self-attested: it carries no third-party timestamp, the blinding is enforced in code, and because the simulation is deterministic given the sealed seed, the seal cannot by itself exclude iteration before sealing.

The generator produced $G = 300$ instances: 112 faithful controls and three *algorithmic* er-

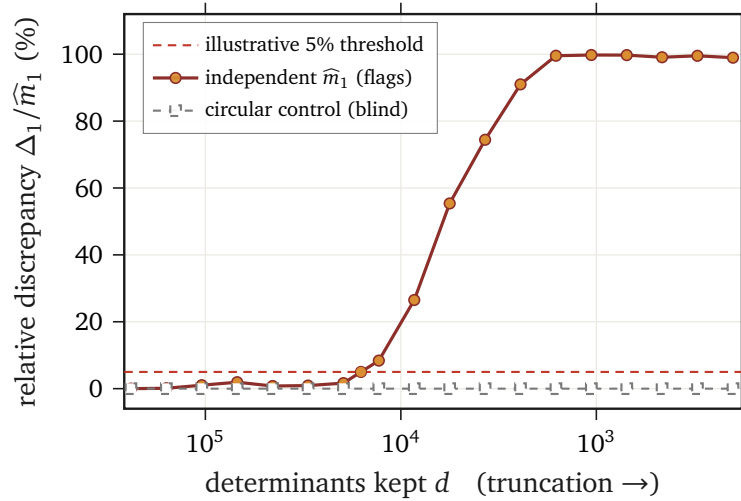


Figure 7: **Independent vs. back-computed moment discrepancy under determinant-subspace truncation (idealized estimator).** The determinant subspace of the doped $L = 12$ Hubbard ring ($N_\uparrow = N_\downarrow = 4$, $U/t = 8$) is truncated to its top d configurations by $|\psi_0|^2$ (the sample-based-diagonalization ordering; full support $n_{\text{sup}} = 2.4 \times 10^5$), and the current first moment $\bar{m}_1$ is recomputed within it. The reconstruction-independent discrepancy $\Delta_1^{\text{rel}} = |\hat{m}_1 - \bar{m}_1|/\hat{m}_1$, with $\hat{m}_1$ on the exact full-sector ground state, stays below 2% down to 8% coverage, crosses an illustrative fixed 5% threshold at $d \approx 1.6 \times 10^4$ (7% coverage) and saturates near 100%. The back-computed (circular) control compares $\bar{m}_1$ with itself and is zero by definition [15, 16]. The screen catches severe truncation, not mild undersampling. Figure 8 removes the full-sector idealization.

ror causes, determinant-subspace (SQD) truncation (56), under-converged Krylov reconstruction (71), and an analytic-continuation artifact adding a spurious spectral feature (61); the classifier saw no labels. The calibrated scope is narrow: every instance uses the current probe J of the doped $L = 6$ Hubbard ring ($N_\uparrow = N_\downarrow = 2$, sector dimension 225) at $U/t \in \{3, 4, 5, 6\}$. Truncations keep the top $d = 29\text{--}138$ of the 186 ground-state configurations by $|\psi_0|^2$ (16–74% of the support) and are mild by weight (75.1–98.7% of $|\psi_0|^2$); in four, $J|0_S\rangle$ has no component inside $\mathcal{S}$ and the reconstruction is empty. Krylov instances are n_l -node Lanczos representations, $n_l \in \{1, 2, 3, 4\}$. A spurious feature is a point atom of weight 2–25% of m_0 at a frequency uniform between the sector’s lowest and highest excitation energies (the pre-registration text calls it a Gaussian atom; the generator adds a δ atom); with probability 1/2 the measure is rescaled to restore m_0 , and in this m_0 -preserving subclass the atom carries 2–20% of the reconstructed weight. The independent estimate is the exact full-sector moment plus unbiased Gaussian shot noise of variance $\text{Var}(O_k)/N_s$ (the idealized estimator of Fig. 7), drawn independently of $\bar{m}_k$, so the test has no same-sample covariance (a joint-sampling simulation finds a same-sample correlation but no power gain; Sec. 9).

The sealed primary endpoint was the pooled detection rate, $\text{TPR} = 101/188 = 53.7\%$ (Wilson 95% interval $[46.6, 60.7]\%$), at a false-positive rate $\text{FPR} = 1/112 = 0.9\%$ ($[0.2, 4.9]\%$). The pooled rate is low because 71 of the 188 corrupted instances are under-converged Krylov reconstructions, which the battery misses (0/71). All 56 truncations were caught ($[93.6, 100]\%$), and 74% of the spurious features (45/61): 93% (25/27) of the weight-changing ones, which the diagonal m_0 channel can reach, and 59% (20/34, $[42, 74]\%$) of the m_0 -preserving ones, which only the off-diagonal moments can reach (a failed sealed prediction; see below). The truncation result is a property of the sampled cuts, not a guar-

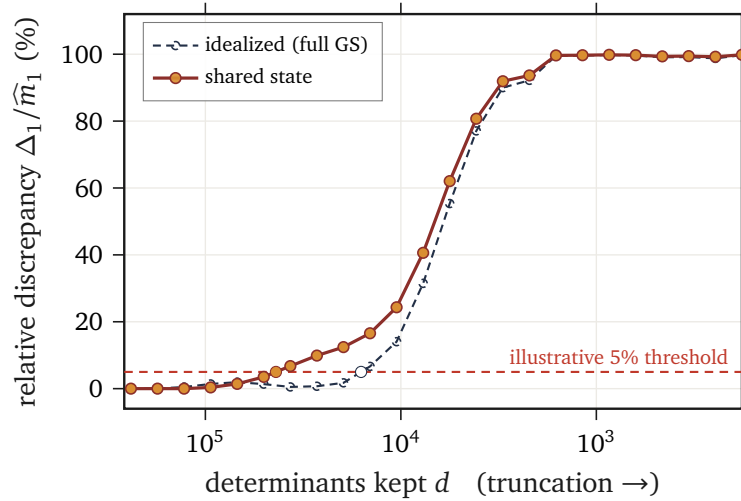


Figure 8: **Shared-state discrimination on one subspace state.** Both first-moment branches use the *same* subspace ground state $|0_S\rangle$ of $H_S = \Pi_S H \Pi_S$, with $e_0 = \langle 0_S | H | 0_S \rangle$ (Appendix B); the evaluation is classical. Solid (oxblood): $\Delta_1^{\text{shared}} = |\hat{m}_1 - \bar{m}_1| / \hat{m}_1$, with $\hat{m}_1 = \langle 0_S | J(H - e_0)J | 0_S \rangle$ from the *full* operators and $\bar{m}_1$ the within- S reconstruction moment; their difference is the $k=1$ leak of Eq. (B.3). Dashed (indigo): the idealized curve of Fig. 7. As d falls (truncation $\rightarrow$), the shared-state discrepancy rises by about five orders of magnitude (non-monotone near both ends), crossing the illustrative 5% threshold at $d \approx 4.4 \times 10^4$ (18% coverage), against 1.6×10^4 (7%) for the idealized curve (accidental dip near $d \sim 4 \times 10^4$). Both saturate near 100%. Doped $L = 12$ ring ($N_\uparrow = N_\downarrow = 4$, $2/3$ filling), $U/t = 8$.

ante: an exhaustive scan of the $U/t = 4$ ring, with exact moments, a deterministic tie order and the same thresholds (Appendix B), finds 12 truncated reconstructions that pass the full battery, three at mid coverage ($d = 86, 87, 95$ of 186) and nine near-complete ($d \geq 160$), plus three narrow windows in which m_1 alone, at the shot-only threshold, has power only at the false-positive level. The near-complete cuts lie outside the catalog's range ($d \leq 138$), no $U/t = 4$ instance of the catalog fell on the mid-coverage ones, and the other U values were not scanned.

The Krylov misses follow from Gauss quadrature: an n_l -node Lanczos representation reproduces $m_0, \dots, m_{2n_l-1}$ exactly [69, 71], and, as the companion study proves, a subspace containing the order- K Krylov space reproduces the moments through order $2K + 1$ [20]. For $n_l \geq 2$ the reconstruction matches m_0 – m_2 to below 10^{-15} relative and is indistinguishable from a faithful one at this order. For $n_l = 1$ it misses m_2 by 2.4–4.2% and passes only because the threshold budgets a 2% bias (largest residual $0.99 \tau_2$). At the shot-only threshold 6 of the 15 one-node instances would fire on m_2 , but so would 5 of the 56 exact $n_l \geq 2$ instances, from noise alone.

Two sealed secondary predictions failed. The one-node Krylov case, expected to be caught, was caught 0/15. The m_0 -preserving spurious-feature subclass, sealed as designed-blind with low expected detection, was caught 20/34, because the rescaling restores only m_0 and still shifts m_1 and m_2 . We report this 59% as an outcome the pre-registration did not anticipate, not as a designed capability; the register was not revised. The other secondary predictions held: high detection of truncations (56/56) and weight-changing artifacts (25/27), low detection of $n_l \geq 2$ Krylov reconstructions (0/56).

The false-positive rate is low by construction: the simulated estimator is unbiased, while the frozen threshold budgets a 2% bias comparable to the shot term. Redrawing the noise

for the 112 faithful instances under the frozen rule gives a modeled FPR of 1.3%, far below the pre-registered nominal target $1 - 0.95^3 = 14.3\%$; one or fewer false positives occur in more than half of the redraws, so the sealed 1/112 is typical. A realized bias of +2% (−2%) on all three moments gives a modeled FPR of 8.0% (11.5%; the threshold scales with $|\widehat{m}_k|$). The 0.9% therefore confirms the implementation, not robustness to real mitigation bias, which remains untested.

Every rejection came from the moment comparisons. The sealed rule describes an interval-Hankel condition, but the classifier checks positivity of the 2×2 Hankel matrix H_1 of the reconstruction moments $\bar{m}_0, \bar{m}_1, \bar{m}_2$, which cannot fail on a catalog of positive measures (smallest eigenvalue over the 300 instances -3.6×10^{-15} , rounding error): its 0/300 holds by construction.

Post hoc, we rescored the 56 sealed truncations with the shared-state estimator of Fig. 8, after a rebuild that reproduces every sealed reconstruction moment, variance and noise draw to rounding error (the rebuild depends on the stored eigensolver tie order inside degenerate $|\psi_0|^2$ shells). With $\widehat{m}_k = \langle 0_S | J(H - e_0)^k J | 0_S \rangle$ plus the sealed noise draw and variances, the frozen rule rejects all 56, every one on m_0 (smallest $r_0 = 5.9$, where $r_k = |\widehat{m}_k - \bar{m}_k|/\tau_k$); before noise, $\widehat{m}_0 - \bar{m}_0 = \widehat{m}_0(1 - w)$ is exactly the weight of $J|0_S\rangle$ outside $\mathcal{S}$. The idealized sealed estimator fired m_0 on 55 of the 56; the exception, id 162 ($r_0 = 0.62$, caught only off-diagonally), is an artifact of the oracle-centered estimate and has $r_0 = 15.8$ under the shared-state estimator. That estimate is itself far from the exact moment (id 162: $\widehat{m}_0 = 1.04$ against $m_0 = 0.37$): the test registers the leak, not the distance to the truth.

Within this catalog the screen detects support truncation and most added spectral features, and misses reconstructions that reproduce the tested moments within the thresholds. The ground truth is classically known and merely withheld from the classifier: this is a blinded test, not a field catch of an error whose correctness is unknown, and no device is in the loop; both remain future work.

Figure 9 shows how the two channels divide the work. Determinant truncation moves the total-weight sum rule $m_0 = \langle J^2 \rangle$, so the diagonal channel catches it (55/56 idealized, 56/56 shared-state), as it catches most weight-changing spurious features (21 of the 25 caught). The m_0 -preserving subclass keeps the total weight and the determinant support fixed by construction, so neither the m_0 residual nor a support-coverage check can see it (cf. Fig. 5); only the off-diagonal moments reject it (20/34, a post hoc reading, since low detection was pre-registered), for example id 222 with $(r_0, r_1, r_2) = (0.16, 3.3, 9.1)$. In all, 26 sealed instances have $r_0 < 1$ but an off-diagonal residual above threshold (m_1 on 16, m_2 alone on 10): the 20 above, 4 weight-changing artifacts whose weight shift fell below τ_0 , truncation id 162 and false positive id 184. The legend counts plotted points: its m_0 -preserving (20) and m_0 -changing (4) entries count only the shaded region, and the other 14 preserving and 23 changing artifacts form “spurious, unshaded” (37). Theorem 2 does not apply: the artifacts add a pole, leaving its fixed-pole class. This off-diagonal reach holds so far only in simulation; the noise analysis of Sec. 6 defers it to lower-error hardware, and on `ibm_fez` only the sampled support came from the device, and the coverage residual was evaluated on it with exact classical amplitudes.

6 Coverage residual on a device-sampled support (IBM Heron)

No moment in this paper is estimated from device counts. On hardware the device supplies only the sampled support $\mathcal{S}$; the moments $\bar{m}_0, \bar{m}_1$ read off the reconstruction are computed classically on $\mathcal{S}$, and the reference values $\widehat{m}_0, \widehat{m}_1$ used below are exact classical numbers. Same-sample estimation, in which $\widehat{m}_k$ and $\bar{m}_k$ come from one set of shots, has been realized only in simulation (Sec. 9).

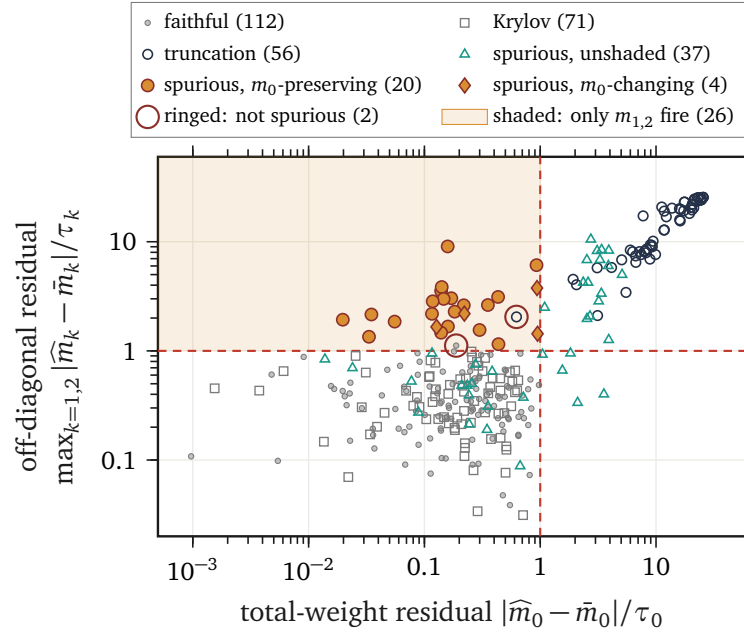


Figure 9: **The diagonal and off-diagonal moments have complementary reach** (sealed, pre-registered blinded battery, doped $L = 6$ ring; SHA-256 0e4c3368... , self-attested). $r_k = |\widehat{m}_k - \bar{m}_k|/\tau_k$ with the frozen thresholds ($z = 1.96$, $N_s = 5 \times 10^4$, assumed 2% bias); $r_k > 1$ is a fire. All 300 plotted at true values. Truncations: 56/56 caught, $r_0 > 1$ in 55. Shaded: 26 rejected only off-diagonally: 20 m_0 -preserving and 4 m_0 -changing spurious features, false positive id 184 and truncation id 162 (both ringed; id 162 has shared-state $r_0 = 15.8$). Calling the m_0 -preserving catches (20/34) a capability is post hoc; low detection was pre-registered. Krylov (0/71): caught by neither channel. Thresholds omit device gate-error bias; on current hardware the simulated off-diagonal moments would carry ~ 37 –116% depolarizing bias (Sec. 6). A structural separation in simulation, not device-realizable firing.

468 The companion study ran a sample-based-diagonalization pipeline on `ibm_fez` at
 469 $L = 6$ (12 qubits; 3.5×10^5 shots, 5×10^4 on each of seven circuits; job
 470 `d9s16avpems73ct6g8g`) [20]. Its retained support is the whole addition sector ($|S| =$
 471 $300/300$), so the reconstruction equals exact diagonalization by coverage (Fig. 11). The bit-
 472 strings were post-selected after being read in reversed qubit order, which swaps the spin labels
 473 and moves the circuits' own output into the discarded $S_z = -1$ sector, so all 300 retained de-
 474 terminants came from device errors [20]. The raw counts are not deposited. The run shows
 475 only that the pipeline executes; no moment residual can be nonzero on a complete support.

476 To evaluate the coverage residual on an incomplete support we ran our own job at $L = 8$
 477 (16 qubits, periodic chain, $U/t = 4$) and deposit its raw counts; we raised the size because at
 478 $L = 6$ even noiseless sampling of the companion's circuits covers the whole sector at 5×10^4
 479 shots per circuit [20]. The reference state $|\varphi\rangle = c_{0\uparrow}^\dagger |0\rangle$, with $|0\rangle$ the exact half-filled ground
 480 state ($E_0 = -4.6035 t$), lies in the addition sector $N_\uparrow = 5$, $N_\downarrow = 4$ of dimension 3920. Its weight
 481 is $m_0 = \langle c c^\dagger \rangle = 1 - \langle n_{0\uparrow} \rangle = 1/2$ exactly, because translation invariance fixes $\langle n_{j\uparrow} \rangle = 4/8$ on
 482 every site (operator and Lehmann values agree to 6×10^{-17}). The seven circuits belong to
 483 the companion's family [20]: each prepares a computational-basis determinant with X gates
 484 (five up and four down electrons) and applies three Trotter steps of on-site R_{zz} and nearest-
 485 neighbor $R_{xx}R_{yy}$ hopping, without Jordan–Wigner strings, up to times 0, 0.5, ..., 3.0 in units
 486 of $1/t$. They conserve particle number and S_z , so without noise they sample only the addition

sector; they do not apply the Hubbard propagator to $c_p^\dagger|0\rangle$ and are not targeted at $|\varphi\rangle$. We ran them at 50, 30, 16 and 4×10^3 shots per circuit (3.5×10^5 in total at the largest budget) with SamplerV2, using measurement twirling, Pauli gate twirling and XpXm dynamical decoupling; no readout-error mitigation was applied (SamplerV2 returns twirled raw counts). Each pooled record is read in the correct bit order and post-selected to the sector, with no configuration recovery, which keeps 23.5–24.1% of the shots (82,697 of 350,000 at 50k: 92% of the shots of the $t = 0$ circuit and 11–13% of each of the six evolved circuits).

The retained bitstrings define $\mathcal{S}$, on which the coverage residual $\Delta_0 = |m_0 - \bar{m}_0| = \langle \varphi | (1 - \Pi_{\mathcal{S}}) | \varphi \rangle$ [Eq. (B.3) at $k = 0$] is evaluated exactly. As the budget falls from 50k to 4k, the configuration coverage $|\mathcal{S}|/3920$ falls from 94% to 35% and the residual grows: $\Delta_0 = 0.0044$, 0.0323, 0.1232 and 0.3726 (Fig. 10). In the companion’s notation $\Delta_0 = m_0(1 - w)$, with w the captured weight of $\mathcal{S}$ [20]; coverage and captured weight differ, and at 50k 94% of the configurations carry 99.1% of the weight ($1 - w = 0.009$). The first-moment residual $\Delta_1 = |\widehat{m}_1 - \bar{m}_1|$ (0.019, 0.057, 0.024, 0.869) is not monotone in the budget; like Δ_0 , it combines a device-sampled support with exact classical amplitudes.

Reference distributions. We compare the device with four reference samplers and report empirical percentile ranks (the fraction of replicas at or below the device value), not Gaussian z -scores, because the weight of $|\varphi\rangle$ is concentrated (1% of the configurations carry 58% of m_0) and the distributions of Δ_0 are skewed.

(i) *Noiseless sampling at the device’s raw shots.* Two thousand multinomial replicas of the exact output distributions of the same seven circuits give $\Delta_0 = 0.036 \pm 0.018$, 0.066 ± 0.027 , 0.121 ± 0.035 and 0.269 ± 0.032 (mean $\pm$ sd) at 50/30/16/4k. The device lies below all 2000 replicas at 50k, at the 1.7th percentile at 30k, at the 57.9th at 16k and above all 2000 at 4k. Its support is larger than in every replica at 50k and 30k ($|\mathcal{S}| = 3682$ against 3379 ± 14 ; 3330 against 3183 ± 16) and smaller than in every replica at 16k and 4k (2798 against 2877 ± 18 ; 1379 against 2027 ± 21). Noiseless sampling cannot reach $\Delta_0 = 0$: the seven output distributions together cover 3864 of the 3920 configurations, leaving 8.8×10^{-5} at infinite shots. The device’s Δ_1 lies at the 35th, 82nd and 27.5th percentiles at 50/30/16k and above all replicas at 4k.

(ii) *Noiseless sampling at the device’s retained counts* puts the device below every replica at 50, 30 and 16k under both count allocations, and at the 0.6th or 5.8th percentile at 4k, with a margin that shrinks as the budget falls (Appendix F).

(iii) *Uniform noise* mixed into the circuits’ output with fraction p lowers the expected Δ_0 at 50k from 0.036 at $p = 0$ to 0.0024 at $p = 1$; at the p that reproduces the device’s retention the mixture’s mean lies below the device at every budget, although the device lies inside the mixture’s replica distribution at 50–16k (Appendix F).

(iv) *Uniform in-sector sampler.* Drawing configurations uniformly from the sector at the device’s retained count N , with no device and no circuit, gives $\mathbb{E}[\Delta_0] = m_0(1 - 1/3920)^N = 3.4 \times 10^{-10}$, 1.5×10^{-6} , 5.2×10^{-4} and 0.093, below the device at every budget. At 50k and 30k this sampler covers essentially the whole sector, and in all ten replicas checked its reconstruction is exact to the Lanczos floor (relative L_1 error below 4×10^{-9}).

What the device run shows. Δ_0 rewards support spreading, not fidelity. The noisy device leaves less weight uncovered than all 2000 noiseless replicas of its own circuits at 50k and than 98% of them at 30k, where its support is the larger one; at 16k and 4k, where its support is smaller, it is typical or worse. Uniform noise lowers Δ_0 at 50k, and a sampler with no quantum content does best, because the 3920-configuration sector is small enough for uniform sampling to cover all or most of it at these budgets (81% of the configurations at 4k); a related effect has been reported for ground-state quantum-selected configuration interaction, where sampling noise can enlarge the configuration space beyond what the ideal ansatz supports [58]. Δ_0 grows as the budget falls for every sampler considered. The run thus shows that the coverage

residual can be evaluated on a device-sampled support; it cannot distinguish a good device from a noisy one and does not certify the reconstruction. Relative to the companion's captured-weight results [20] (Sec. 1), this section adds the evaluation on incomplete, device-sampled supports at four budgets with deposited counts, and the reference distributions above.

A device-data false negative. The relative L_1 error of each reconstruction, $\int |A_S - A| d\omega / \int A d\omega$ on the job's 600-point frequency grid at $\eta = 0.15 t$, is 0.35, 0.56, 0.90 and 1.12 at 50/30/16/4k. At 50k both residuals, $\Delta_0 = 0.0044$ and $\Delta_1 = 0.019$, lie below the bias terms that the pre-registered rule of Sec. 5 would set ($\tau_k \geq z b m_k$ with $z = 1.96$ and $b = 2\%$: $\tau_0 \geq 0.020$, $\tau_1 \geq 0.082$), yet the relative L_1 error is 0.35. These thresholds are counterfactual: the rule was sealed for the $L = 6$ current probe, and here $\widehat{m}_0, \widehat{m}_1$ are exact. This is consistent with the companion's result that captured weight alone cannot certify a reconstruction: its bound places the error above $1 - w = 0.009$ and needs the boundary coupling for an upper bound [20]. In L_1 error the device beats all 80 noiseless replicas at 50k and loses to all of them at 4k, and the uniform in-sector sampler beats the device at every budget (Appendix F, which also records a superseded noise model and a nonparametric bootstrap, neither used as a test).

The discriminating off-diagonal estimator (m_1, m_2 from rotated, qubit-wise-commuting bond-basis settings, Fig. 12) is verified only in simulation (Appendix B); on present hardware, at the companion's circuit depth, its forecast bias is a large fraction of the signal. Global depolarizing noise pulls the estimate of $m_1 = \langle 0|O|0 \rangle$, $O = c_{0\uparrow}(H - E_0)c_{0\uparrow}^\dagger$, toward $\text{Tr}(O)/\dim \mathcal{H}$, which for the $L = 8$ device Hamiltonian is 3.24 times m_1 , so the bias is upward (Appendix B). An exact density-matrix simulation of a bond-basis probe with nearly the same ratio (the $q = \pi$ density probe of the open $L = 6$ chain at $U/t = 4$: $m_1 = 10.6$, $\text{Tr} / \dim = 34.3$, ratio 3.23) with local two-qubit depolarizing noise at $\varepsilon = 2.5 \times 10^{-3}$ gives a first-moment bias of 13% of the signal at 100 CZ and 37% at 292 CZ, the companion's depth; this static, local noise model omits the operator spreading of a real circuit and is a lower bound. Global depolarizing at the same depths gives 49% and 116%, and a coherent-ZZ model 47% at 292 CZ. The particle-number post-selection that protects the diagonal channel does not survive the bond-basis rotations. A device-side off-diagonal catch therefore awaits lower gate error; no device datum here bears on off-diagonal content (Table 2).

The off-diagonal limitation concerns the bond-basis estimator of m_1 , not every discriminating on-device channel. A diagonal one exists: the momentum distribution $n_k = \int_{\omega < 0} A(k, \omega) d\omega = \langle c_{k\sigma}^\dagger c_{k\sigma} \rangle$, the occupied spectral weight at momentum k . The total weight $\int A(k, \omega) d\omega = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$ is pinned at every k and is blind to how the weight splits between the occupied and empty sides; n_k is not. Read out diagonally after a number-conserving fermionic Fourier transform, n_k keeps the particle-number post-selection that the bond-basis rotations lose. It is not immune to noise: depolarization flattens it toward the same filling value that a dispersion corruption does, so it discriminates only corruptions whose flattening exceeds the device's own (forecast and scope in Appendix F).

A pre-registered on-device n_k test was sealed on 2026-09-02 after a simulated Stage-0 gate passed, with little margin, at $T_1 \geq 150 \mu\text{s}$ and $T_2 \geq 100 \mu\text{s}$. Its single permitted amendment would have relaxed these floors to $T_1 = 100 \mu\text{s}$, $T_2 = 70 \mu\text{s}$; the worst-admissible row (two-qubit error parameter $\varepsilon = 3.5 \times 10^{-3}$; the forecasts above assume 2.5×10^{-3}) failed its sealed simulated gate, with upper confidence bound 0.0287 against the sealed threshold 0.0267. A row identical except for $\varepsilon = 2.85 \times 10^{-3}$ passed, but ε is a compound knob (depolarizing and coherent-ZZ components together), not a physical two-qubit error rate. The amendment was denied and the experiment has not been run (Appendix F); like the off-diagonal catch, it awaits lower-error hardware.

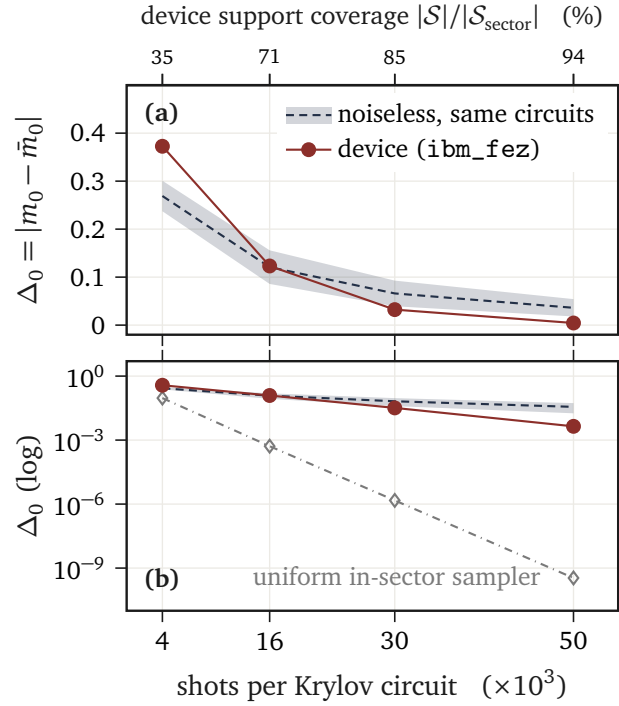


Figure 10: **Coverage residual on device-sampled supports** (ibm_fez, $L = 8, 16$ qubits, $U/t = 4$, sector dimension 3920, seven circuits). Oxblood: Δ_0 , exact on the device-retained support $\mathcal{S}$; top axis of (a): coverage $|\mathcal{S}|/3920$. Indigo: 2000 noiseless multinomial replicas of the same circuits at the device’s raw shots per circuit (dashed, mean; shading, ± 1 sd, right-skewed, not a 68% interval). Grey diamonds, (b) only: uniform in-sector sampler at the device’s retained count N , $\mathbb{E}[\Delta_0] = m_0(1 - 1/3920)^N$. (a) Linear, (b) log scale. The device lies below all noiseless replicas at 50k and above all at 4k (percentiles in text); the uniform sampler, which uses no device, scores lower at every budget. Δ_0 measures support coverage and rewards support spreading; it is not a fidelity diagnostic.

7 A closed-form bound: the Christoffel miss-distance

We now state what a *passing* screen implies. The bound below is classical (moments and orthogonal polynomials); our contribution is its evaluation as a diagnostic for a reconstructed spectrum, not the mathematics, and the Christoffel–Darboux kernel is an established data-analysis tool [69, 70, 98, 99]. Let $F(t) = \int_{-\infty}^t A(\omega) d\omega$ be the cumulative weight of a diagonal $T = 0$ Lehmann measure whose moments $m_0, \dots, m_{2n}$ are known exactly. They fix the orthonormal polynomials $p_0, \dots, p_n$ via the Hankel system [65], and the Christoffel function is $W_n(t) \equiv \lambda_n(t) = 1/K_n(t, t) = 1/\sum_{j=0}^n p_j(t)^2$, with $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ the Christoffel–Darboux kernel (construction from the moments in Appendix C) [69, 70].

Theorem 1 (Cumulative-weight miss-distance bound (classical; re-derived)). *Let $d\mu = A(\omega) d\omega$ be a positive measure of compact support (a diagonal $T = 0$ Lehmann spectrum) whose Hankel matrix $H_n = [m_{i+j}]_{i,j=0}^n$ is positive definite, and suppose the moments $m_0, \dots, m_{2n}$ are known exactly. For $t \in \mathbb{R}$ define the cumulative-weight miss-distance as the extremal spread $\sup |F_{\nu_1}(t) - F_{\nu_2}(t)|$ over all positive measures ν_1, ν_2 on $\mathbb{R}$ sharing those moments. Then for every $t \in \mathbb{R}$ this spread equals the Christoffel function $W_n(t)$, which is also the supremum of the mass a moment-matching measure can place at t (the Gauss–Radau weight $\lambda_n(t)$, attained except at the n zeros of the orthonormal polynomial p_n). In particular, any reconstruction whose*

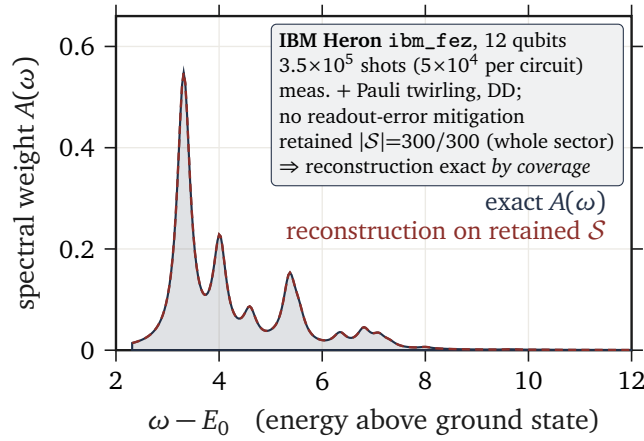


Figure 11: **Executability run of the companion study ($L = 6$, full-sector support).** Exact $A(\omega)$ (indigo) and reconstruction on the device-retained support (dashed); $L = 6$ Hubbard model, $U/t = 4$, addition sector $4 \uparrow, 3 \downarrow$ (300 configurations), $\eta = 0.15 t$. `ibm_fez`, 12 qubits, 3.5×10^5 shots (5×10^4 on each of seven circuits; job `d9s16avpemt73ct6g8g`); measurement twirling, Pauli gate twirling and dynamical decoupling, no readout-error mitigation [20]. The retained support is the whole sector ($|\mathcal{S}| = 300/300$), so the reconstruction equals exact diagonalization by coverage. The bitstrings were read in reversed qubit order before post-selection, so all 300 retained determinants came from device errors. The run shows only that the pipeline executes; it says nothing about accuracy at incomplete coverage (Fig. 10).

602 moments $m_0, \dots, m_{2n}$ equal the true ones obeys $|F_{\text{rec}}(t) - F_{\text{true}}(t)| \leq W_n(t)$ at every $t \in \mathbb{R}$
 603 (Appendix C) [100, 101].

604 *Remarks.* No convex-hull hypothesis is needed: for $H_n > 0$ the Gauss–Radau rule with a node
 605 at t has positive weights for every real t except the n zeros of p_n , and its atom at t is $W_n(t)$
 606 (Appendix C). The bound is monotone, $W_{n+1} \leq W_n$. For N distinct poles, $H_n > 0$ only for
 607 $n \leq N - 1$, and at a pole $W_n(\omega_i) \geq w_i > 0$ (the residue), with equality at $n = N - 1$: W_n does
 608 not vanish on the support. Pointwise vanishing $W_n(t) \rightarrow 0$ is a thermodynamic-limit statement
 609 that needs regularity of the limiting measure (Appendix C) [83, 97].

610 The equality is the classical content of the Chebyshev–Markov–Stieltjes inequalities and
 611 Gauss–Christoffel moment theory [65, 66, 69]. We say “bound”, not “certify”: the screen never
 612 certifies that $A(\omega)$ is correct. The bounded object is cumulative weight, not the pointwise value
 613 of $A(\omega)$ or a peak position [69], and the contraction *rate* holds only under that regularity [83,
 614 97]. The top panel of Fig. 13 shows the mechanism: the true $F(t)$ lies between the extremal
 615 Gauss–Radau envelopes, a bracket of width $W_n(t)$.

616 The theorem assumes exact moments; the deployed screen passes a reconstruction whose
 617 residuals lie within tolerances, $|\Delta_k| \leq \tau_k$, and only through order $2n = 2$. For the measure
 618 of Fig. 13 this has two consequences. First, at $n = 1$ the exact-moment bracket is the one-
 619 sided Chebyshev (Cantelli) bound, $W_1(\omega) = m_0 \sigma^2 / [\sigma^2 + (\omega - \bar{\omega})^2]$ with mean $\bar{\omega} = 11.10 t$
 620 and variance $\sigma^2 = 2.10 t^2$; its maximum m_0 lies at the mean, $0.01 t$ from the dominant atom
 621 ($11.09 t$), so the bracket is vacuous at the peak, and just above the top pole, at $\omega = 14 t$, it is
 622 still 0.20 . Second, tolerances loosen it further: the largest atom that a positive measure with
 623 $|m_k - m_k^{\text{true}}| \leq \tau_k$ ($k \leq 2$, the deployed threshold rule evaluated on this state) can place at ω is
 624 3.2 – 7.7 times W_1 at $\omega = 14$ – $20 t$ (0.63 against $W_1 = 0.20$ at $14 t$), 10 – 11 times W_1 far outside
 625 the band ($\omega \approx 34$ – $50 t$), and more than m_0 at the mean ($1.045 m_0$). At exact-moment orders
 626 $n \leq 4$ ($m_0, \dots, m_8$) the global maximum falls only to $\max_t W_4 = 0.882$, and at $n = N - 1 = 4$,
 627 W_4 at the dominant atom equals its residue 0.778 and stays there (panel c). No positive-

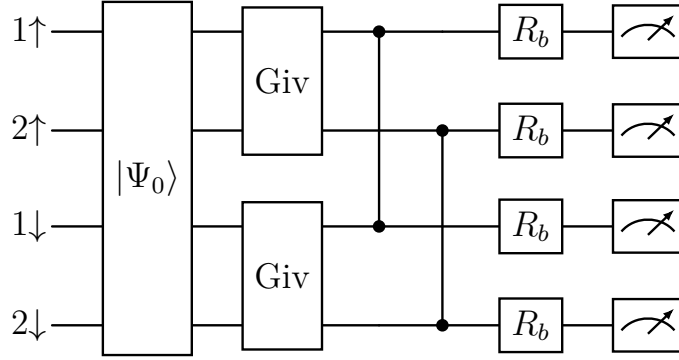


Figure 12: **Measurement circuit of the bond-basis moment estimator (schematic; not run on hardware).** One circuit, drawn for a two-site unit cell (4 qubits: two orbitals per spin). The first blocks stand for state preparation: reference state $|\Psi_0\rangle$, number-conserving hopping per spin species (Giv), on-site interaction $U\hat{n}_\uparrow\hat{n}_\downarrow$ (controlled up–down pairs). A per-qubit rotation R_b (H or $S^\dagger H$ for X or Y , identity for Z) precedes a computational-basis measurement; each qubit-wise-commuting Pauli group needs one circuit. The all- Z setting gives m_0 ; the hopping content of m_1, m_2 needs rotated settings. The verification of Appendix B samples the exact ground state in each basis instead of running a preparation. On present hardware the off-diagonal estimate would carry the bias of Sec. 6.

order moment beyond m_2 is estimated in this work. At the deployed order, passing the screen therefore places no constraint on the cumulative weight at the peak. This is the moment-level counterpart of the companion’s finding that captured weight alone cannot certify a sample-based spectral function [20].

Refutation and bracketing are separate statements. Rejection ($|\Delta_k| > \tau_k$ for some k) does not involve W_n , and its reach is set by the thresholds (Sec. 8); a Hankel/Stieltjes violation of the *estimated* moments rejects the estimate, not a non-negative reconstruction. W_n governs only how tightly a *passing* reconstruction’s cumulative weight is bracketed. The looseness above therefore limits what passing implies; it does not limit the power to reject.

At a frequency the measure leaves empty, however, $W_n(t)$ bounds the weight any moment-matching reconstruction can place there; far above the band ($\omega \approx 34\text{--}50t$), $W_1 \sim 10^{-3}$ and the higher orders are far smaller (Appendix C). These small values reflect the distance from the band, not resolution: at $n = 1$ they are the Cantelli tail. Nearer the band the bracket is wider ($W_1 = 0.20$ at $\omega = 14t$), and the deployed tolerances raise the far-out-of-band $n = 1$ values about tenfold. This is a computed property of this measure, not a theorem, and we do not extrapolate it to the two-sided single-particle Mott gap, a distinct measure not tested here.

Proposition 1 (Two-sided Markov–Krein window-mass bound). *Let $d\mu = A(\omega)d\omega$ be a positive measure (a diagonal $T = 0$ Lehmann spectrum) with $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$, and suppose $m_0, \dots, m_{2n}$ are known exactly. For any frequency window $[c, d]$ with $c \leq d$ real, the mass $\mu([c, d]) = \int_c^d A(\omega)d\omega$ of every positive measure sharing those moments obeys*

$$\begin{aligned} \max\{0, F_-(d) - F_+(c)\} &\leq \mu([c, d]) \\ &\leq \min\{m_0, F_+(d) - F_-(c)\}, \end{aligned} \tag{1}$$

where $F_\pm(t)$ are the lower/upper principal (Gauss–Radau) cumulative representations of Lemma 3 (App. C): $F_-(t)$ is the mass that the $(n+1)$ -point Gauss–Radau rule with a node fixed at t places

strictly left of t , and $F_+(t) = F_-(t) + \lambda_n(t)$ with $\lambda_n(t) = W_n(t)$ its atom at t (at the n zeros of p_n , where that rule degenerates, both are taken as limits). The bracket is two-sided. Its width is at most $W_n(c) + W_n(d)$ (with equality before intersecting with the trivial bounds $[0, m_0]$), so it inherits the monotone contraction $W_{n+1} \leq W_n$ and is tight wherever both endpoints fall in weight-poor frequency. The sharp Krein–Nudelman extremum, attained by a principal representation with prescribed nodes at c and d (equivalently, the optimum of the scalar truncated-moment problem over positive measures), is no wider. Equation (1) follows directly from the Chebyshev–Markov–Stieltjes inequalities and extends the one-sided point bound $|F_{\text{rec}} - F_{\text{true}}| \leq W_n$ to a two-sided bound on integrated weight [67, 69, 100, 101].

Evaluated on the same $L = 6$, $U/t = 8$ current measure, again at exact moments, the two-sided bracket (1) is tight when both window edges lie in weight-free frequency (a peak-containing window gives $\mu \in [0.9732, 0.9758]$ around the exact 0.9744 at $n = 4$, and $[0.848, 1]$ at the deployed order) and as loose as W_n when an edge sits on the peak; the deployed tolerances loosen it further (Appendix C). It is the real-axis, ground-state-moment counterpart of the Euclidean-time two-sided spectral bounds obtained from moment-problem semidefinite programs [78–80]; its moments are ground-state operator expectations, not lattice correlators. As with the one-sided bound, we do not extrapolate it to the two-sided single-particle Mott gap.

8 Detectability and its sample-complexity limit

The Christoffel bound of Sec. 7 concerns what passing implies; rejection has a matching characterization: which errors the moment battery detects in exact arithmetic, and what it guarantees at finite thresholds, where the Vandermonde structure of the moment map sets the shot cost. The limiting error class is the *within-sector redistribution*: a corruption that keeps the retained determinant support (so the leakage estimator of Sec. 4 is blind by construction) and only reweights the reconstructed poles. The statement below is not a new theorem: part (a) restates the classical Prony/Vandermonde uniqueness of an atomic measure from its moments [65, 102], and part (b) turns it, with an elementary shifted-power bound, into a detectability-versus-shot-budget statement.

Theorem 2 (Detectability and completeness of the moment battery). *Let $A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n)$ be a diagonal $T = 0$ Lehmann measure on N distinct poles $\{\omega_n\}$ with $w_n = |\langle n|J|0\rangle|^2 \geq 0$, and let a within-sector redistribution $\mathbf{w} \rightarrow \mathbf{w} + \boldsymbol{\delta}$ ($\boldsymbol{\delta} \neq 0$, keeping $w_n + \delta_n \geq 0$) act on the same poles. Write $\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$, the induced shift $\Delta_k = |(V\boldsymbol{\delta})_k| = |\sum_n \delta_n \omega_n^k|$, and $\rho = \|\boldsymbol{\delta}\|_1$. Then $\|\boldsymbol{\delta}\|_2 \leq \rho \leq \sqrt{N} \|\boldsymbol{\delta}\|_2$, and $\rho/2$ is the weight moved when m_0 is preserved.*

(a) (Completeness in exact arithmetic.) *If the redistribution preserves $m_0, \dots, m_K$ with $K < N - 1$, then $\Delta_j > 0$ for at least one j with $K < j \leq N - 1$; quantitatively*

$$\max_{K < j \leq N-1} \Delta_j \geq \frac{\sigma_{\min}(V_{0:N-1})}{\sqrt{N-1-K}} \|\boldsymbol{\delta}\|_2, \quad (2)$$

with $V_{0:N-1}$ the $N \times N$ node-Vandermonde on rows $0, \dots, N-1$ and both sides in one fixed frequency unit. Hence the order- $(N-1)$ battery is complete for fixed-pole redistributions: no nonzero within-sector redistribution preserves all of $m_0, \dots, m_{N-1}$. If the poles may also move (two measures of at most N atoms each), completeness holds at order $2N-1$ by Prony counting [65, 102]. The only alteration invisible to every moment is weight traded among degenerate eigenstates at one frequency, which leaves $A(\omega)$ unchanged and is not an error.

(b) (Finite thresholds.) Let each moment carry the noise floor $\delta_k = \sqrt{\text{Var}(O_k)/N_s + b_k^2}$ at N_s shots, with bias budget b_k , and the two-sided threshold $\tau_k = z_{1-\alpha/2} \delta_k$. We call a given δ exposed at order j when its noise-free shift exceeds the threshold, $\Delta_j > \tau_j$; whether a finite-shot test then rejects it is a question of power (Corollary below). Some $m_0, \dots, m_K$ -preserving redistribution is exposable at order $K+1$ iff $\Delta_{K+1}^{\max} > \tau_{K+1}$, with $\Delta_{K+1}^{\max}$ the linear program (E.1). If rows $0, \dots, K+1$ of V have a nontrivial null space ($N - K - 2 \geq 1$), the smallest Δ_{K+1} over such redistributions is zero for every size ρ up to a limit set by positivity: the guaranteed power at order $K+1$ is zero, however well separated the poles. Two conditions bracket detection.

(i) Sufficient. If δ preserves $m_0, \dots, m_K$, it is exposed at some order $j \leq N - 1$ whenever the right-hand side of (2) exceeds $\max_{K < j \leq N-1} \tau_j$.

(ii) Necessary (cluster bound). If δ preserves $m_0, \dots, m_{s-2}$ and is supported on poles within an interval of diameter D and midpoint ω_c , then $(V\delta)_{s-1} = \sum_n \delta_n (\omega_n - \omega_c)^{s-1}$, so $\Delta_{s-1} \leq \rho (D/2)^{s-1}$; every higher shift obeys $\Delta_j \leq \rho \sum_{i=s-1}^j \binom{j}{i} |\omega_c|^{j-i} (D/2)^i$, which is also $O(\rho D^{s-1})$. Exposure at order $s-1$ therefore requires

$$\rho > \frac{\tau_{s-1}}{(D/2)^{s-1}}, \quad \text{equivalently} \quad (3)$$

$$N_s > \frac{z_{1-\alpha/2}^2 \text{Var}(O_{s-1})}{\rho^2 (D/2)^{2(s-1)} - z_{1-\alpha/2}^2 b_{s-1}^2},$$

and no N_s suffices when $\rho (D/2)^{s-1} \leq z_{1-\alpha/2} b_{s-1}$. The shot cost to detect a redistribution of fixed size at its first exposable order grows at least as $D^{-2(s-1)}$ as the cluster contracts.

Proof. Appendix E.

Remarks. Super-resolution bounds for clustered nodes [103–106] are proved for Fourier nodes, and part (b)(ii) does not rely on them; for real nodes the same $\sigma_{\min} \propto D^{s-1}$ scaling, at fixed relative geometry of the cluster, follows from Gautschi's bound [107, 108], so for an isolated cluster the sufficient condition (i) weakens with the same power as the necessary bound (ii) (Appendix E, which also gives the shot-cost scaling under uniform cluster contraction). The fixed-pole class excludes under-converged Krylov reconstructions, whose poles move; their misses are explained in Sec. 5.

Corollary (detection power vs. cluster size and shot count). Under the Gaussian estimator validated in Fig. 14 (skewness -0.04 , excess kurtosis -0.05), the test reaches power $1 - \beta$ at level α when $\Delta_j \gtrsim z_{\alpha,\beta} \delta_j$, with $z_{\alpha,\beta} = z_{1-\alpha/2} + z_{1-\beta}$ and the bias budget counted in δ_j as in the deployed rule. By part (b)(ii) this requires $N_s \gtrsim z_{\alpha,\beta}^2 \text{Var}(O_{s-1}) / [\rho^2 (D/2)^{2(s-1)} - z_{\alpha,\beta}^2 b_{s-1}^2]$ at the first exposable order, and no budget suffices below the bias-limited cluster diameter $D_* = 2(z_{\alpha,\beta} b_{s-1} / \rho)^{1/(s-1)}$. Without bias the cost is finite for every $D > 0$ and diverges only as $D \rightarrow 0$.

For the deployed $L = 6$, $U/t = 4$ battery the $N = 5$ poles are well separated (minimal separation $\Delta\omega_{\min} = 0.87 t$; on nodes rescaled to $[-1, 1]$, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$), yet its guaranteed power against fixed-pole redistribution is zero. The reason is the null-space dimension, not conditioning: rows $0, 1, 2$ of V leave a two-dimensional null space on five poles. Over m_0, m_1 -preserving redistributions the linear program (E.1) puts the signed shift of m_2 in $[-0.92, 7.09]$ (Table 3): up to $3.8 \tau_2$ ($\tau_2 = 1.848$), but the range contains zero. Three explicit redistributions pass the deployed battery. CE1, in the null space, preserves m_0, m_1, m_2 exactly, moves 72% of the weight ($\rho = 1.43 m_0$), empties two of the five poles and cuts the dominant residue from 0.393 to 0.037. CE2 preserves m_0, m_1 and sets $\Delta_2 = \tau_2$ while moving 86% of the weight (dominant residue $0.393 \rightarrow 0.073$). A third, CE3, also passes, as does the random

redistribution committed with the original analysis (Appendix E). Orders 3 and 4 would catch some of these, but not all: with the same rule ($\tau_3 = 15.0$, $\tau_4 = 122$; the local-estimator variance behind them is a lower bound), CE2 and CE3 give $\Delta_3 \approx 3.3 \tau_3$ and $\Delta_4 \approx 7.1\text{--}7.2 \tau_4$, whereas CE1 gives $0.23 \tau_3$ and $0.70 \tau_4$; over all redistributions that keep m_0, m_1, m_2 exactly, $\Delta_3 \leq 0.35 \tau_3$, and Δ_4 reaches $1.19 \tau_4$ at the one that maximizes Δ_3 , a marginal case. Neither m_3 nor m_4 is estimated in this work. The residual blind spot thus has two sources: the battery order, since with $m_0, \dots, m_K$ every fixed-pole redistribution in the $(N-K-1)$ -dimensional null space passes, regardless of pole separation; and pole clustering, which by (3) raises the shot cost of detection at least as $D^{-2(s-1)}$. Injectivity guarantees that some m_j separates any fixed-pole redistribution; the thresholds decide whether that separation is observable at a given budget.

9 Discussion and limitations

This paper contributes a reconstruction-independent moment screen for sample-based quantum spectral functions; estimators that do not read the moments off $A(\omega)$ (classical full-operator evaluation on the sampled subspace state, or bond-basis measurements on a prepared ground state, simulated only; Appendix B); a self-attested, hash-sealed blinded calibration where the truth is known; and classical bounds on what a pass implies. The screen is a necessary condition: failing it rejects the reconstruction at the pre-registered level (per-moment $\alpha = 0.05$; blinded false-positive rate 0.9%, 1/112, low by construction for an unbiased simulated estimator; Sec. 5), and passing it excludes only the tested failure modes. Independence from $A(\omega)$ is what allows the two branches to disagree; it does not turn agreement into a statement about the line shape.

Relation to the companion study. The companion [20] already contains the captured weight ($\Delta_0 = m_0(1-w)$), the moment-exactness theorem that explains the Krylov passes, the leakage identity [Eq. (B.3) at $k = 2$], and the result that captured weight alone cannot certify, of which the 50k run of Sec. 6 ($\Delta_0 = 0.0044$, relative L_1 error 0.35 at $\eta = 0.15 t$) is a device-data instance. What this paper adds is stated in Sec. 1.

Positioning by mechanism. Self-verifying variational simulation certifies a prepared state through an energy-variance bound measured on the device [41]; observable-estimation validation certifies error-mitigated *expectation values* by validating the noise model and the mitigation [52]. Our screen certifies neither: it tests a reconstructed spectral *function* for weight misassignment, documented by the companion for sample-based spectral functions [20] and, for subspace methods in general, our extrapolation from the limitations documented for their ground-state energies [54–58]. Cross-checking the classical diagonalizer across independent LAPACK paths [109] catches solver errors but not sampling incompleteness or weight misassignment. The physical approaches are complementary: a hardware spectrum could be produced under observable-estimation validation from a self-verified input state and then screened by our moments. We claim only the third role.

Constructive alternatives. Moment information can instead be built *into* the reconstruction: Nevanlinna continuation imposes the Pick/positivity structure of a causal response [110], maximum-entropy and stochastic continuation take moments and sum rules as prior information [63, 64], and Prony, minimal-pole and stochastic-pole-expansion fits can pin low-order moments to their operator values [111, 112]. Where available this is preferable, but such a spectrum is consistent only with the moments *it* was given, within the method that enforced them, for the data it was built from. Irmejs and Santos propose building dynamical correlation functions on hardware without real-time evolution, from weight- $O(k)$ moments measured on a prepared eigenstate, prove exponential accuracy away from the real axis, and demonstrate

the method in classical simulations of small lattices [113]; a circuit-differentiation framework builds the retarded Green's function from derivatives of real-time-evolution circuits under realistic noise [114]. These programs construct. Our screen is their diagnostic counterpart for a consumer handed a finished $A(\omega)$ by a pipeline it does not control, on the real axis, where the constructive guarantees are weakest. Building the moments in and screening the result are two layers of one defense, not alternatives.

The Christoffel bound (Sec. 7) bounds cumulative weight, not pointwise $A(\omega)$ or peak positions; its contraction rate needs regularity [83, 97]; at the deployed order it is vacuous at the spectral mean, and a reconstruction passing with tolerances is bracketed only by a tolerance-inflated extremum ($3.2\text{--}7.7 W_1$ at $\omega = 14\text{--}20 t$ on the measure of Fig. 13). A passing screen does not resolve a line shape [69, 70, 74].

The limits of the screen are the following.

Necessary, not sufficient; zero guaranteed power at the deployed order. Infinitely many measures share any finite set of moments, so passing can never certify correctness [65, 66]. On the five weighted poles of the $L = 6$, $U/t = 4$ current measure, m_0, m_1, m_2 leave a two-dimensional null space, so the guaranteed power against fixed-pole redistributions is zero whatever the pole separation, because of the null-space dimension $N - K - 1$, not conditioning. Redistributions moving 72% of the weight ($m_0\text{--}m_2$ kept) and 86% (m_0, m_1 kept, $|\Delta_2| \leq \tau_2$) both pass (Sec. 8). The blind spot closes at order $N - 1$ in exact arithmetic (Theorem 2), which needs moments beyond the deployed order (up to m_4 here), not estimated in this work; m_3 at the deployed shot budget would not close it, since no redistribution keeping m_0, m_1, m_2 exact shifts it by more than $0.35 \tau_3$. Truncations also escape: at a 2% bias the full battery passes 12 of 185 truncated reconstructions of the $U/t = 4$ ring (Appendix B). Device data show the same limitation: the 50k $L = 8$ reconstruction has $\Delta_0 = 0.0044$ and a relative L_1 error of 0.35 (Sec. 6).

Error classes. The screen addresses weight and subspace misassignment; the Gauss-law check addresses gauge-symmetry violation at the state level. Neither is a general noise diagnostic; device-noise trust remains the domain of observable-estimation validation [52]. Reconstructions that preserve the estimated moments pass. Under-converged Krylov reconstructions are the clearest case (0/71 in the blinded test): with two or more Lanczos nodes they match $m_0\text{--}m_2$ exactly, as the companion's moment-exactness theorem predicts [20], and the one-node case passes only within the 2% bias budget (Sec. 5).

Positivity. The positivity that underwrites the moment bounds is that of a diagonal $T = 0$ Lehmann measure. Off-diagonal and finite-temperature spectral functions are not positive measures in general and need separate treatment [38].

Idealized estimator in the calibration. The blinded test simulated the estimate as the exact full-sector moment plus unbiased Gaussian shot noise, with no same-sample covariance and no mitigation bias, while the threshold budgets a 2% bias. Its 0.9% false-positive rate is therefore low by construction: the modeled rate is 1.3% against the sealed nominal joint level of 14.3%, and a realized +2% (−2%) bias raises it to 8.0% (11.5%) (Sec. 5). Robustness to a real mitigation bias is untested. The post hoc shared-state rescoring also rejects all 56 truncations, and the estimator form had to be fixed in advance, because its operator forms differ off an eigenstate by enough to change a verdict (Appendix A).

Same-sample statistics. Strict same-shot estimation of both branches has been realized only in simulation, with a local estimator that uses exact ground-state amplitudes and is not device-realizable. One set of shots gives both $\hat{m}_k$ and $\bar{m}_k$ with correlation $\rho \approx 0.43$ at 84% coverage; treating them as independent inputs, as one would for an interval from separate data [82], makes the test over-conservative (realized false-positive rate 1.5% at a nominal 5%), and at matched false-positive rate the correlation-aware test gains no detection power (Appendix B).

Hardware. The device supplies only the sampled support; no moment has been estimated from device counts. The coverage residual rewards support spreading rather than fidelity: at

50k shots per circuit the device lies below all 2000 noiseless-sampling replicas of its circuits (at 4k above all), and a uniform in-sector sampler scores lower at every budget. The discriminating moments m_1, m_2 remain simulation-only: the depolarizing bias of the bond-basis estimate of m_1 is at least 37% of the signal at the companion circuit depth and two-qubit error 2.5×10^{-3} (a lower bound), and the sealed simulated gate for a diagonal n_k run failed at the relaxed noise floors of its single permitted amendment (Sec. 6).

Synthetic errors and the seal. Every error rejected here is synthetic: an injected corruption or a controlled truncation of otherwise exact data. The labels are withheld from the classifier, but the truth is classically known and the calibrated scope is narrow: the current probe of the doped $L = 6$ ring at $U/t \in \{3, 4, 5, 6\}$, truncations keeping 15.6–74.2% of the support configurations but 75.1–98.7% of the ground-state weight, and spurious point atoms carrying 2–25% of m_0 . The seal is self-attested: it has no third-party timestamp, the blinding is implemented in code, and since the simulation is deterministic given the sealed seed, the seal cannot by itself exclude iteration before sealing. No error has yet been caught in a spectrum whose correctness was unknown in advance; that field test is future work.

Scale. One-dimensional Hubbard rings at $L \leq 12$ are solved outright by density-matrix renormalization group (DMRG) and exact diagonalization, so no quantum resource is needed for their spectra. This is deliberate: calibration requires exact moments, and the screen is characterized against these methods, not in competition with them. Its intended regime, where no controlled classical spectral reference exists, is real-time dynamics of two-dimensional and frustrated lattices and *ab initio* molecular spectra. A code-scalability check at $L = 24$ (open chain, half filling, $U/t = 8$, $S_z = 0$ sector of dimension $\binom{24}{12}^2 = 7.31 \times 10^{12}$) takes $\mathcal{S}$ from Born-rule samples of a TeNPy DMRG ground state [115] and evaluates both branches on the same subspace ground state. From 500 to 15 000 shots Δ_0 falls from 44.2 to 24.5 (DMRG $m_0 = 55.0$), tracking the DMRG-confirmed reconstruction error; even at 15 000 shots the subspace energy $e_0 = -2.90 t$ is far above the DMRG $E_0 = -7.66 t$, so the firing is expected. The check shows that the code runs at this scale; it is not a measure of power and not a beyond-classical claim.

Several extensions follow. The interval-moment formulation is demonstrated on the exact state (Appendix B); a device-side deployment needs the mitigation bias measured and bounded rather than assumed, for example from error-model violation [116] or by validating the noise model and the mitigation [52]. Device estimation of the discriminating moments awaits lower gate error; the pre-registered n_k device run was not executed because its amendment failed the sealed simulated gate (Sec. 6). The Gauss-law check, evaluated here exactly rather than from samples, shows that any exactly known operator identity can serve as a test, inviting a catalog of conservation laws, symmetry sectors and sum rules, each screening a different error class [39, 74]. Extending positivity-based bounds to off-diagonal and finite-temperature spectra, where Lehmann positivity fails, is the main theoretical opening [38]. Built on kernel-polynomial, recursion and quadrature machinery [69, 74, 75], the screen fits the sample-based diagonalization and Krylov pipelines that generate the spectra it tests [15–17, 20, 117, 118].

9.1 The verification stack: two necessary layers

Evaluated as full-operator expectation values on the sampled subspace state, the battery is classical post-processing; a device-native estimate of m_1, m_2 needs separate bond-basis settings on a prepared ground state (Appendix B; simulated only). The estimator branch is the reconstruction-free counterpart of classical shadows [50, 119], whose spectral specialization extracts gaps without forming $A(\omega)$ [120] and whose fermionic variant constructs dynamical correlation functions [121]. Those methods construct; our screen tests a finished $A(\omega)$ it did not produce.

The moment vector of every positive measure lies in the moment body: its Hankel matrices, and its localizing matrices on any interval known to contain the support, are positive semidefinite [65, 66]. A violated Hankel condition on the estimated moments rejects the estimate without reference to $A(\omega)$ (Fig. 2); it cannot reject a non-negative reconstruction, and applied to the reconstruction moments in the blinded test it fired on none of the 300 instances.

Wang *et al.* turn the same machinery into certified two-sided bounds on many-body ground-state properties [81], and Mortimer *et al.* extend it to finite measurement statistics [82]. Applied to the ground-state operators O_k , these semidefinite programs (SDPs) would bound the true moments, $m_k \in [\ell_k, u_k]$; this is possible in principle but not done in the cited works or here. Membership of the read-off moments $\bar{m}_k$ in such intervals would be a second necessary condition, of the same logical type as ours; their conjunction is at least as strict as either alone and is not a composition with a sufficient condition. No finite set of moment conditions certifies $A(\omega)$.

The classical moment-body conditions (not the Wang/Mortimer programs) illustrate such bounds at exact moments on the $L = 6$, $U/t = 8$ current measure normalized to $m_0 = 1$ (true $m_2 = 125.4$). Positivity gives $m_2 \geq m_1^2/m_0 = 123.3$. The Hausdorff condition $\int (\omega - a)(b - \omega) d\mu \geq 0$, for a support contained in $[a, b]$, gives $m_2 \leq (a+b)m_1 - ab m_0$: 147.6 for the weighted support $[2.49, 13.92]t$, which is oracle knowledge requiring the exact spectrum; a priori, 243.3 for the excitation range $[0, 21.91]t$ of the probe's $N = 4$, $S_z = 0$ sector and 579.8 for the full Fock range $[0, 52.21]t$ (573.7 with the lower edge at the smallest positive Fock excitation, $0.148t$). An under-dispersed moment vector ($m_2 = 122.1$) violates the positivity bound; an over-dispersed one ($m_2 = 148.8$) violates only the oracle Hausdorff bound and passes both a-priori bounds. Neither condition has been applied to device-estimated moments.

The two layers could meet on integrated window weight: over all positive measures that share given moments, the window mass $\mu([c, d]) = \int_c^d A d\omega$ ranges over an interval, given in closed form at exact moments by Proposition 1 (the Markov-Krein envelope, of width at most $W_n(c) + W_n(d)$), and an SDP fed with interval-valued moments [82] would give a finite-statistics counterpart against which a reconstruction's window mass could be checked. We do not implement this. Nor do we attempt frequency-resolved weights, a limitation rather than a stylistic choice: a weight $|\langle n|J|0 \rangle|^2$ at a resolved frequency is an excited-state matrix element, outside the ground-state moment hierarchy that makes these SDPs tractable; the resolved line shape almost certainly remains beyond this machinery.

10 Conclusions

We have developed and calibrated, building on the companion's captured-weight and moment-exactness results [20], a reconstruction-independent, necessary-condition screen for sample-based quantum spectral functions. It compares the low-order moments read off a reconstructed $A(\omega)$ with ground-state operator expectations evaluated without reference to $A(\omega)$. A failed comparison rejects the reconstruction; a passed one excludes only the tested failure modes. In a self-attested, hash-sealed blinded simulation of 300 instances on one doped $L = 6$ ring the pre-registered primary endpoint was a detection rate of 53.7% (101/188; Wilson 95% interval $[46.6, 60.7]\%$) at a false-positive rate of 0.9% (1/112; $[0.2, 4.9]\%$), with an unbiased simulated estimator, so the false-positive rate is low by construction. The screen caught every determinant truncation in the test set (56/56; an exhaustive depth scan of the $U/t = 4$ ring finds 12 of 185 that pass) and 74% of the spurious features (45/61). It caught no under-converged Krylov reconstruction (0/71), and two sealed secondary predictions failed. At the deployed order m_0 – m_2 its guaranteed power against fixed-pole weight redistributions is zero; in exact arithmetic this blind spot closes only at order $N - 1$ for N poles.

The hardware run shows less. On IBM `ibm_fez` the device supplied only the sampled support of an $L = 8$ electron-addition sector. The coverage residual Δ_0 grows from 0.004 to 0.37 as support coverage falls from 94% to 35%, but it rewards support spreading rather than device fidelity: at 50k shots per circuit the device lies below every noiseless-sampling replica of the same circuits, a uniform in-sector sampler does better at every budget, and the 50k reconstruction has a relative L_1 error of 0.35 while its residual is small. The run shows that the diagonal channel executes on device data. It does not validate the device, the reconstruction or the discriminating moments.

Several problems remain open: estimating m_1 and m_2 on hardware, where the depolarizing bias of the bond-basis estimate of m_1 is at least 37% of the signal at the companion circuit depth and a two-qubit error of 2.5×10^{-3} , and the diagonal n_k alternative did not pass its sealed simulated gate; calibrating the screen with a realistic mitigation bias and a same-sample estimator realized on a device (so far realized only in simulation); closing the within-sector blind spot with higher moments at an affordable shot cost; applying the semidefinite moment bounds of Refs. [81, 82] to the same moment sequence; and a field test on a spectrum whose correctness is not known in advance.

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Data availability The code and data that support the findings of this study are archived on Zenodo (DOI to be added at submission) and openly available in the GitHub repository <https://github.com/nicolasbonilla/reconstruction-independent-moment-tests>. The repository contains the exact sector-Lanczos/Haydock engine; the interval-moment, blinded-test, within-sector and hardware-analysis scripts; the pre-registration with its SHA-256 seal and the sealed blinded records; the per-figure data; and Makefile targets and a notebook that run the generators. The file `docs/REPRODUCE.md` maps each figure and the key quoted numbers to their scripts and data files, and lists the items that do not yet regenerate from source. The raw measurement counts of our four $L = 8$ `ibm_fez` jobs are included (jobs `da983qse74ec73ajfr2g`, `da98k16rbfbs73chckvg`, `da98icsjbipc73fej78g` and `da98cksjbipc73fej1tg` at 50, 30, 16 and 4×10^3 shots per circuit), together with the reference-distribution generator `src/delta0_reference_mc.py` (output `data/2026-09-27_delta0_reference_mc.json`) and the exporter `src/export_device_dat.py` of Fig. 10. The raw counts of the companion's $L = 6$ run (Fig. 11) are not deposited; that run was post-selected in reversed bit order, so all of its retained determinants came from device errors [20]. The sealed blinded files are the canonical record: regenerating the truncation instances from the seed depends on tie-breaking inside degenerate $|\psi_0\rangle^2$

shells, and the tie order that reproduces them is stored with the blinded-test addenda (data/2026-09-27_blind_addenda.json).

A Derivation of the moment identities

We derive the operator form of the spectral moments used throughout. Each m_k is a ground-state expectation value computable without the dynamics, and we state where the derivation needs an exact eigenstate. Let $H = H^\dagger$ have a non-degenerate ground state $|0\rangle$, $H|0\rangle = E_0|0\rangle$, and a complete orthonormal eigenbasis $\{|n\rangle\}$ with $H|n\rangle = E_n|n\rangle$; let $J = J^\dagger$ be a Hermitian probe (a current or density). The diagonal zero-temperature Lehmann spectral function

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0)) \quad (\text{A.1})$$

is a non-negative measure supported on $\omega = E_n - E_0 \geq 0$, and its power moments are $m_k = \int \omega^k A(\omega) d\omega = \sum_n (E_n - E_0)^k |\langle n|J|0\rangle|^2$.

Writing $|\langle n|J|0\rangle|^2 = \langle 0|J|n\rangle \langle n|J|0\rangle$, using $\langle n|(H - E_0)^k = (E_n - E_0)^k \langle n|$, and closing with completeness $\sum_n |n\rangle \langle n| = \mathbb{1}$,

$$\begin{aligned} m_k &= \sum_n \langle 0|J|n\rangle (E_n - E_0)^k \langle n|J|0\rangle \\ &= \langle 0|J (H - E_0)^k J|0\rangle. \end{aligned} \quad (\text{A.2})$$

The ground state is annihilated by $H - E_0$, so acting to the right $(H - E_0)J|0\rangle = [H, J]|0\rangle$, and iterating,

$$(H - E_0)^k J|0\rangle = \text{ad}_H^k(J)|0\rangle, \quad \text{ad}_H(J) \equiv [H, J], \quad (\text{A.3})$$

where ad_H^k denotes the k -fold nested commutator $[H, [H, \dots [H, J]]]$. Substituting Eq. (A.3) into Eq. (A.2) gives the closed operator form

$$m_k = \langle 0|J \text{ad}_H^k(J)|0\rangle \equiv \langle O_k \rangle, \quad (\text{A.4})$$

with $O_k = J \text{ad}_H^k(J)$ built solely from H and J —no reference to the spectrum or to the reconstructed $A(\omega)$.

The lowest moments specialize immediately. For $k = 0$, Eq. (A.4) gives the total weight $m_0 = \langle 0|J^2|0\rangle = \langle J^2 \rangle$. For $k = 1$, $m_1 = \langle 0|J[H, J]|0\rangle$; expanding the symmetric double commutator $[J, [H, J]] = 2JHJ - J^2H - HJ^2$ and using $H|0\rangle = E_0|0\rangle$,

$$\langle 0|[J, [H, J]]|0\rangle = 2\langle 0|J(H - E_0)J|0\rangle = 2m_1, \quad (\text{A.5})$$

so that

$$m_1 = \frac{1}{2} \langle 0|[J, [H, J]]|0\rangle. \quad (\text{A.6})$$

This is the first-moment (double-commutator) sum rule of A_{JJ} ; for the current operator J it fixes the mean frequency of the spectral weight. It is distinct from the optical-conductivity f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$, which is the m_{-1} moment of A_{JJ} (a ground-state kinetic energy) [35–37]. On an exact eigenstate the symmetrized form (A.6) is manifestly real and independent of the additive constant E_0 , which the one-sided expression $\langle 0|J[H, J]|0\rangle$ obscures.

Negative-order moment. The negative-order moment $m_{-1} = \int_0^\infty A_J(\omega)/\omega d\omega$ of Fig. 5 carries the kernel $1/\omega$ and weights low frequencies; on the ring of that figure it rejects after m_1 and m_2 . This is the optical f -sum, and it too is a ground-state expectation: since the

current is the polarization velocity $J = i[H, P]$, $m_{-1} = \frac{1}{2}\langle [P, [H, P]] \rangle = \frac{1}{2}\langle -\hat{T} \rangle$, a kinetic energy (equivalently $\int_0^\infty \text{Re } \sigma_{\text{reg}} d\omega = \pi m_{-1}$). On open chains we verify this identity against the Lehmann sum over the computed poles, to a relative 2×10^{-15} at $L = 6$ and 8 and 2×10^{-14} at $L = 12$ [35–37]. On the ring the polarization is multivalued and the identity acquires Kohn’s stiffness correction, $m_{-1} = \frac{1}{2}\langle -\hat{T} \rangle - \mathcal{D}$ with $\mathcal{D} = \frac{1}{2} \partial_\theta^2 E_0$ the finite-size Drude weight (verified at $L = 12$ to a relative 2×10^{-6} , with $\mathcal{D}$ from the flux curvature). The open chain ($\mathcal{D} = 0$) does not restore an advantage for the corruption of Fig. 5, because 41% of its regular weight already lies below $6t$ and $m_{-1} = 4.81$ is large: there the thresholds are crossed at $f \approx 0.17$ for m_{-1} , 0.059 for m_1 and 0.045 for m_2 . When the spurious pole is moved across $0.25\text{--}4t$, m_{-1} rejects first only for poles at $0.25t$ and $0.5t$, on both geometries (at $0.25t$ on the ring, $f \approx 0.015$ against 0.044 for m_1 and 0.043 for m_2). On open chains m_{-1} is also cheap: $\langle -\hat{T} \rangle$ is a one-body operator, measured in two qubit-wise-commuting settings that are a subset of those the m_0, m_1 estimator already requires (Appendix B).

Off an eigenstate the operator forms differ. Equations (A.2), (A.4) and (A.6) agree because $|0\rangle$ is an exact eigenstate of H . On any other state $|\psi\rangle$, for example the ground state $|0_S\rangle$ of a truncated subspace Hamiltonian (Appendix B), the form $\langle \psi | J(H - e_0)^k J | \psi \rangle$ with $e_0 = \langle \psi | H | \psi \rangle$ and the commutator form $\langle \psi | J \text{ad}_H^k(J) | \psi \rangle$ are no longer equal for $k \geq 1$ (for $k = 1$ the real part of the latter is $\frac{1}{2}\langle \psi | [J, [H, J]] | \psi \rangle$), and in general neither equals the exact moment m_k . The difference can decide a verdict. Take the current probe of the doped $L = 6$, $U/t = 4$ ring and truncate its sector to the $d = 100$ determinants of largest $|\psi_0|^2$ (44% of the 225-dimensional sector, 54% of the 186-determinant ground-state support). With the determinants ordered deterministically (descending $|\psi_0|^2$, ties broken by sector index, NumPy `lexsort`), the subspace ground state gives $\langle J(H - e_0)J \rangle = 4.558$ but $\frac{1}{2}\langle [J, [H, J]] \rangle = 3.717$, against the reconstruction moment $\bar{m}_1 = 3.743$ and the exact $m_1 = 4.227$. At the deployed threshold $\tau_1 = 0.229$ (Appendix B) the first form rejects the truncated reconstruction, which is the correct verdict since $\bar{m}_1$ is 11% below the exact value, and the second does not. This cut falls inside a 24-fold degenerate shell of $|\psi_0|^2$, so these values depend on how ties are broken: NumPy `argsort`, which does not fix the order of ties, gave 4.605, 3.707 and 3.810 on our machine, with the same two verdicts (data/2026-09-27_theory_numerics.json, script src/estimator_form_offeigenstate.py). We therefore fix the estimator as $\hat{m}_k = \langle \psi | J(H - e_0)^k J | \psi \rangle$ with $e_0 = \langle \psi | H | \psi \rangle$ on the prepared state, and use the commutator forms only on exact eigenstates.

Two features of Eq. (A.4) make it the operational core of the screen. First, O_k is an explicit operator polynomial in H and J , a weighted sum of computational-basis observables, so $\langle O_k \rangle$ can be evaluated on a prepared state without forming $A(\omega)$; this is the estimator $\hat{m}_k$ of Appendix B. Besides the idealized evaluation on the exact ground state, it is evaluated here in three ways, none on hardware: classically on the subspace state that also yields the reconstruction (Fig. 8, a deterministic $|\psi_0|^2$ truncation; the $L = 24$ DMRG test of Sec. 9, a shot-sampled subspace); as a local estimator from the same simulated shots, which needs the exact amplitudes (joint sampling, Appendix B); and from simulated Pauli-grouped measurements of a separately prepared ground state (Appendix B). Second, the derivation used only completeness of the eigenbasis and that $|0\rangle$ is an exact eigenstate, so it holds for any Hermitian probe and Hamiltonian, which is why the screen transports across models. Positivity of the measure (A.1) is not needed for the identities; it enters only in the positivity conditions and the bounds built on them.

For a non-Hermitian probe the same steps hold with $J^\dagger$ on the bra side, $m_k = \langle 0 | J^\dagger (H - E_0)^k J | 0 \rangle$, so that $m_0 = \langle J^\dagger J \rangle$. The single-particle spectral function $A_{k\sigma}(\omega)$ of Fig. 3 is the two-sided (photoemission plus inverse-photoemission) spectrum of the fermionic probe $c_{k\sigma}$,

and its moments close under *anticommutators* rather than the double commutator:

$$\begin{aligned} \int A_{k\sigma} d\omega &= \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1, \\ \int \omega A_{k\sigma} d\omega &= \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle = \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A.7})$$

the Hartree-shifted band centroid [62]. The Hermitian double-commutator map above is the special case $J = J^\dagger$; Eq. (A.7) is the version used for the single-particle screen of Fig. 3, and it is likewise a ground-state expectation of explicit operators.

Collective probes. For the non-Hermitian probes ρ_q and S_q^z of Fig. 6 the double commutator is the symmetrized form $\frac{1}{2}[m_1(q) + m_1(-q)]$, equal to $m_1(q)$ by the inversion symmetry of the ring; both forms are evaluated on the exact ground state, where they coincide. Only the kinetic term contributes to m_1 , since the Hubbard interaction is diagonal in the occupation basis and commutes with ρ_q and S_q^z .

Derivation of the anticommutator moment identities. Equation (A.7) is not a fresh sum rule but the non-Hermitian specialization of Eqs. (A.2)–(A.4), derived here for completeness. Write $d \equiv c_{k\sigma}$, $d^\dagger \equiv c_{k\sigma}^\dagger$, with $\{|n\rangle\}$ the complete eigenbasis of H over all particle-number sectors, $H|n\rangle = E_n|n\rangle$, ground state $|0\rangle$, energy E_0 . The two-sided single-particle spectral function is the sum of an electron-addition ($N+1$) and an electron-removal ($N-1$) branch,

$$\begin{aligned} A_{k\sigma}(\omega) &= \sum_n |\langle n|d^\dagger|0\rangle|^2 \delta(\omega - (E_n - E_0)) \\ &\quad + \sum_n |\langle n|d|0\rangle|^2 \delta(\omega + (E_n - E_0)), \end{aligned} \quad (\text{A.8})$$

a non-negative measure whose support *straddles* $\omega = 0$ (addition poles at $\omega = E_n - E_0 > 0$, removal poles at $\omega = -(E_n - E_0) < 0$). Its power moments split accordingly,

$$\begin{aligned} \mu_p \equiv \int \omega^p A_{k\sigma} d\omega &= \sum_n (E_n - E_0)^p |\langle n|d^\dagger|0\rangle|^2 \\ &\quad + (-1)^p \sum_n (E_n - E_0)^p |\langle n|d|0\rangle|^2. \end{aligned} \quad (\text{A.9})$$

Applying to each branch the steps of Eqs. (A.2)–(A.4)— $(E_n - E_0)^p \langle n|d^\dagger|0\rangle = \langle n|\text{ad}_H^p(d^\dagger)|0\rangle$ and completeness—gives the closed operator form

$$\mu_p = \langle 0|d \text{ad}_H^p(d^\dagger)|0\rangle + (-1)^p \langle 0|d^\dagger \text{ad}_H^p(d)|0\rangle, \quad (\text{A.10})$$

built solely from H , d , and $d^\dagger$. For $p = 0$ this is the canonical anticommutator $\mu_0 = \langle 0|dd^\dagger + d^\dagger d|0\rangle = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$. For $p = 1$, using $H|0\rangle = E_0|0\rangle$ and $\langle 0|H = E_0\langle 0|$ on the outer factors one verifies term by term that

$$\mu_1 = \langle 0|d[H, d^\dagger]|0\rangle - \langle 0|d^\dagger[H, d]|0\rangle = \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle, \quad (\text{A.11})$$

the anticommutator first-moment identity, manifestly real and independent of the additive constant E_0 . For the Hubbard Hamiltonian $H = \sum_{k'\sigma'} \epsilon_{k'} c_{k'\sigma'}^\dagger c_{k'\sigma'} + U \sum_i n_{i\uparrow} n_{i\downarrow}$ the commutator evaluates in closed form: the kinetic term gives $[c_{k\sigma}, H_0] = \epsilon_k c_{k\sigma}$, while $[c_{k\sigma}, H_U] = \frac{U}{\sqrt{L}} \sum_j e^{-ikj} c_{j\sigma} n_{j,-\sigma}$ (with $c_{k\sigma} = \frac{1}{\sqrt{L}} \sum_j e^{-ikj} c_{j\sigma}$), and since $\{c_{j\sigma} n_{j,-\sigma}, c_{j'\sigma}^\dagger\} = \delta_{jj'} n_{j,-\sigma}$,

$$\begin{aligned} \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle &= \epsilon_k \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle + \frac{U}{L} \sum_j \langle n_{j,-\sigma} \rangle \\ &= \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A.12})$$

the Hartree-shifted band centroid quoted in Eq. (A.7) and evaluated per momentum in Fig. 3 [62].

Scope: which checks transport to the single-particle channel. The zeroth- and first-moment checks carry over intact: μ_0 and μ_1 are ground-state expectation values of explicit operators [Eqs. (A.10)–(A.11)], so they give reconstruction-independent residuals Δ_0, Δ_1 at each k . So does the two-sided Hamburger–Hankel positivity $H_n = [\mu_{i+j}] \geq 0$, since a positive measure on $\mathbb{R}$ has a positive semidefinite Hankel matrix whatever the sign of its support; it tests the estimated moments, not a non-negative reconstruction (Appendix B). The shifted-Stieltjes condition $H_n^{(1)} = [\mu_{i+j+1}] \geq 0$ does *not* transport: it presupposes support on $[0, \infty)$, and $A_{k\sigma}$ is supported on both signs of ω . The Christoffel bracket of Sec. 7 applies to any positive measure on $\mathbb{R}$, but we have evaluated it only for the one-sided current measure and do not deploy it here. The single-particle screen is therefore Δ_0, Δ_1 and Hamburger–Hankel positivity; the shifted-Stieltjes condition belongs to the one-sided current and density responses.

B The reconstruction-independent moment estimator

We give the explicit estimator $\widehat{m}_k$ and show why it is independent of the reconstruction and, unlike the spectral moment $\bar{m}_k$, detects subspace truncation. Sample-based diagonalization returns a set $\mathcal{S}$ of computational-basis determinants $\{|x\rangle\}$ and diagonalizes the projected Hamiltonian $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$, with $\Pi_{\mathcal{S}} = \sum_{x \in \mathcal{S}} |x\rangle\langle x|$, on a classical computer [15, 16]. This gives eigenpairs $\{E_n^{\mathcal{S}}, |n_{\mathcal{S}}\rangle\}$, among them the subspace ground state $|0_{\mathcal{S}}\rangle = \sum_{x \in \mathcal{S}} c_x |x\rangle$ with energy $e_0 = \langle 0_{\mathcal{S}} | H | 0_{\mathcal{S}} \rangle$. Unless $\mathcal{S}$ contains the support of the exact ground state, $|0_{\mathcal{S}}\rangle$ is an eigenstate of $H_{\mathcal{S}}$ but in general not of H , so the operator forms of Appendix A differ on it, and we use the one fixed there. The probe J and the Hamiltonian H are sparse in this basis, with known matrix elements $J_{yx} = \langle y | J | x \rangle$ and $H_{yx} = \langle y | H | x \rangle$.

The moment $\langle 0_{\mathcal{S}} | J (H - e_0)^k J | 0_{\mathcal{S}} \rangle$ admits two evaluations that coincide only when $\mathcal{S}$ is complete for the operators involved. Inserting the *subspace* resolution $\Pi_{\mathcal{S}}$ in place of the identity yields the reconstruction (spectral) moment,

$$\begin{aligned} \bar{m}_k &= \sum_{n \in \mathcal{S}} (E_n^{\mathcal{S}} - e_0)^k |\langle n_{\mathcal{S}} | J | 0_{\mathcal{S}} \rangle|^2 \\ &= \langle 0_{\mathcal{S}} | J [\Pi_{\mathcal{S}} (H - e_0) \Pi_{\mathcal{S}}]^k J | 0_{\mathcal{S}} \rangle \\ &= \langle 0_{\mathcal{S}} | J (H_{\mathcal{S}} - e_0)^k J | 0_{\mathcal{S}} \rangle, \end{aligned} \tag{B.1}$$

which is exactly $\int \omega^k A(\omega) d\omega$ and lives entirely inside $\mathcal{S}$. Here $H_{\mathcal{S}} - e_0$ acts on $\text{span } \mathcal{S}$, so the $k = 0$ term carries the convention $[\Pi_{\mathcal{S}} (H - e_0) \Pi_{\mathcal{S}}]^0 \equiv (H_{\mathcal{S}} - e_0)^0 \equiv \Pi_{\mathcal{S}}$, and $\bar{m}_0 = \langle 0_{\mathcal{S}} | J \Pi_{\mathcal{S}} J | 0_{\mathcal{S}} \rangle$ is the weight captured by $\mathcal{S}$ rather than the full $\langle J^2 \rangle$. The reconstruction-independent estimator instead applies the sparse operator $O_k = J (H - e_0)^k J$ directly,

$$\begin{aligned} \widehat{m}_k &= \langle 0_{\mathcal{S}} | J (H - e_0)^k J | 0_{\mathcal{S}} \rangle \\ &= \sum_{x, x' \in \mathcal{S}} c_x^* c_{x'} \langle x | J (H - e_0)^k J | x' \rangle, \end{aligned} \tag{B.2}$$

where each matrix element $\langle x | O_k | x' \rangle$ is a finite product of sparse operators summing over intermediate determinants that need *not* lie in $\mathcal{S}$. This is the operational meaning of independence: $\widehat{m}_k$ probes the action of J and H on $|0_{\mathcal{S}}\rangle$ beyond the sampled subspace, whereas $\bar{m}_k$ is confined to it. Evaluated on $|0_{\mathcal{S}}\rangle$, Eq. (B.2) is the shared-state estimator of Fig. 8. Evaluated instead on the exact ground state $|0\rangle$ (with $e_0 \rightarrow E_0$), it gives the exact moment m_k ; this idealized version, more independent than a device allows, is used in Fig. 7, in the sealed blinded

test (Sec. 5, with Gaussian shot noise added), and in the Monte-Carlo and interval-battery demonstrations below.

The two evaluations agree when the states that H generates from $\varphi \equiv J|0_S\rangle$ lie in $\text{span } \mathcal{S}$: if $\text{span } \mathcal{S}$ contains the Krylov space $\text{span}\{\varphi, H\varphi, \dots, H^K\varphi\}$, then $[\Pi_{\mathcal{S}}(H - e_0)\Pi_{\mathcal{S}}]^j\varphi = (H - e_0)^j\varphi$ for $j \leq K$ and $\widehat{m}_k = \bar{m}_k$ for every $k \leq 2K + 1$. This moment exactness, and its failure at order $2K + 2$, are proved and measured in the companion study [20]. It explains why the under-converged Krylov reconstructions of Sec. 5 pass the low-order battery: an n_l -step Lanczos reconstruction started from the probe state projects onto its Krylov space with $K = n_l - 1$ (the argument holds for any orthogonal projector, not only a determinant subspace) and so reproduces $m_0, \dots, m_{2n_l-1}$. The one-step case reproduces only m_0 and m_1 ; its m_2 is 2–4% low and passes only through the 2% bias budget (Sec. 5). Under determinant-subspace truncation the Krylov space is not contained, and

$$\widehat{m}_k - \bar{m}_k = \langle \varphi | (H - e_0)^k | \varphi \rangle - \langle \varphi | [\Pi_{\mathcal{S}}(H - e_0)\Pi_{\mathcal{S}}]^k | \varphi \rangle \quad (\text{B.3})$$

measures the spectral weight the truncation leaks outside $\mathcal{S}$.

At $k = 0$, with the convention above, Eq. (B.3) reduces to $\Delta_0 = \langle \varphi | (\mathbb{1} - \Pi_{\mathcal{S}}) | \varphi \rangle = \widehat{m}_0(1 - w) \geq 0$, with $w = \|\Pi_{\mathcal{S}}\varphi\|^2/\|\varphi\|^2$ the companion's captured weight ($\Pi_{\mathcal{S}}$ is its $P_{\mathcal{S}}$) [20]; with $\varphi = c_p^\dagger|0\rangle$ built from the exact ground state and only $\mathcal{S}$ taken from the device, it is the coverage residual of Sec. 6. When $w = 1$ the $k = 1$ term vanishes as well, and the first term that can be nonzero is the companion's boundary leakage $\Delta_2 = \|(\mathbb{1} - \Pi_{\mathcal{S}})H\varphi\|^2$. The full-support reconstruction of Fig. 14 is an instance: with $\mathcal{S}$ the whole 186-determinant support of the $L = 6$, $U/t = 4$ ground state, $w = 1$, Δ_0 and Δ_1 vanish to machine precision, and $\Delta_2 = 0.62$ (1.9% of $m_2 = 33.39$); it captures all of φ and is still not exact. These identities are the companion's [20], as are the bounds on the reconstruction error by $1 - w$ from below and by the boundary coupling from above and the result that captured weight alone cannot certify; their use as a thresholded rejection test is this paper's (Sec. 1). For the shared-state estimator, $\Delta_k = |\widehat{m}_k - \bar{m}_k|$ is nonzero only if $\mathcal{S}$ misses part of φ or of the states $H^j\varphi$ that order k probes; the independent estimator can then fire, while the circular control ($\bar{m}_k$ against itself) returns $\Delta_k \equiv 0$ (Sec. 4). Weight moved *within* $\mathcal{S}$ is a separate blind spot (Appendix E).

Equation (B.2) is equivalently an importance-weighted average over $\mathcal{S}$,

$$\widehat{m}_k = \sum_{x \in \mathcal{S}} |c_x|^2 O_k^{\text{loc}}(x), \quad O_k^{\text{loc}}(x) = \frac{\langle x | O_k | 0_S \rangle}{\langle x | 0_S \rangle}, \quad (\text{B.4})$$

the local-operator estimator familiar from variational Monte Carlo [123]; the local moment $O_k^{\text{loc}}(x)$ again sums over determinants connected to x by O_k and so reaches outside $\mathcal{S}$. Drawing N_s configurations from $|c_x|^2$ and averaging O_k^{loc} gives an unbiased Monte-Carlo estimate with variance $\text{Var}(O_k)/N_s$, where $\text{Var}(O_k) = \sum_x |c_x|^2 |O_k^{\text{loc}}(x)|^2 - \widehat{m}_k^2$ is the local-estimator variance. This form needs the amplitudes c_x , which a device does not return, so we use it only in simulation (Fig. 14 and the joint-sampling study below). On a device the same expectation would be estimated from Pauli-grouped measurements, described at the end of this appendix, with a variance of its own.

Same-sample statistics. Strict same-shot estimation of both branches has been realized only in simulation. In the joint-sampling analysis of the $L = 6$, $U/t = 4$ current probe (reproduction materials), one set of shots per replica gives both $\widehat{m}_k$ and $\bar{m}_k$; there $\widehat{m}_k$ is the local estimator above, which uses the exact ground-state amplitudes, so it is not device-realizable. The falsifier variance is $\text{Var}(\Delta_k) = \text{Var}(\widehat{m}_k) + \text{Var}(\bar{m}_k) - 2\text{Cov}(\widehat{m}_k, \bar{m}_k)$, with a same-sample correlation $\rho \approx 0.43$ at 84% coverage that decays towards zero as coverage approaches 100%. Treating the two estimates as independent inputs, as one would when a moment interval comes from separate data, for example from the finite-statistics certificate of Mortimer *et al.* [82], over-estimates this

variance, and the test becomes over-conservative in the undersampled regime (realized false-positive rate 1.5% at a nominal 5%). At matched false-positive rate the correlation-aware test gives no detection-power advantage (power differences between -5.6 and $+0.05$ percentage points across the pre-registered ladder of corruptions and coverages), so the covariance is a calibration correction, not a power gain. This structure is a quantum instance of Hausman-type specification testing [124]: $\widehat{m}_k$ is the consistent, model-free estimator and $\bar{m}_k$ the model-implied one. The independent-input variance $\text{Var}(\widehat{m}_k) + \text{Var}(\bar{m}_k)$ is exact only at full coverage, where $\rho \rightarrow 0$; Hausman's efficiency limit, $\text{Cov}(\widehat{m}_k, \bar{m}_k) = \text{Var}(\bar{m}_k)$, would instead reduce the falsifier variance to $\text{Var}(\widehat{m}_k) - \text{Var}(\bar{m}_k)$, and our correlation lies between the two limits.

The exact-moment test takes its finite-shot form here. We model the estimator uncertainty as $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$, a shot term over N_s samples plus a bias budget b_k for mitigation and device error, and reject the reconstruction at level α when

$$\Delta_k = |\widehat{m}_k - \bar{m}_k| > \tau_k, \quad \tau_k \equiv z_{1-\alpha/2} \delta_k.$$

The deployed rule uses $\alpha = 0.05$ per moment ($z_{0.975} = 1.96$), $N_s = 5 \times 10^4$ and $b_k = 0.02 |\widehat{m}_k|$; for the current probe of the doped $L = 6$, $U/t = 4$ ring it gives $\tau_0 = 0.029$, $\tau_1 = 0.229$ and $\tau_2 = 1.848$. A second, separate check concerns the estimate alone: whether some moment vector in the box $[\widehat{m}_\ell - \tau_\ell, \widehat{m}_\ell + \tau_\ell]$ satisfies the positivity conditions obeyed by the moments of every positive measure, $H_n \geq 0$ and, for one-sided $T = 0$ support, $H_n^{(1)} \geq 0$. Infeasibility would show that the estimated moments belong to no spectrum, pointing to the device, the mitigation or the estimator; it cannot refute a non-negative reconstruction, whose own moments $\bar{m}_k$ satisfy these conditions by construction. In the implementations reported here (the sealed blinded test and the interval battery below) the positivity condition is evaluated at $\bar{m}_k$ and therefore cannot fire; every rejection comes from a residual Δ_k . The operators, targets and intervals of each demonstration are in the reproduction materials.

The estimator also admits a device-native evaluation. We verify it in simulation because it settles how the off-diagonal (hopping) content of $\widehat{m}_1, \widehat{m}_2$ would be obtained on hardware. For a probe diagonal in the computational basis, the density probe $\rho_q = \sum_j \cos(qj) n_j$ at $q = \pi$, the moment operator $O_k = \delta \rho_q (H - E_0)^k \delta \rho_q$ is a sum of Pauli strings, and $\widehat{m}_k = \langle 0 | O_k | 0 \rangle = \sum_P c_P \langle P \rangle$ is a linear combination of Pauli expectations estimated from counts. The m_0 term is a pure computational-basis estimator. The hopping terms in m_1, m_2 need rotated-basis measurements, one circuit per qubit-wise-commuting group of Pauli strings, the standard grouping for Pauli observables; classical shadows are a randomized alternative [50]. This path measures a prepared ground state in rotated bases, with circuits separate from those whose samples define $\mathcal{S}$: it shares the device with the reconstruction but not the samples.

We verify it by finite-shot sampling of the exact ground state of an open $L = 6$ Hubbard chain at $U/t = 4$ (12 qubits), with 5×10^4 shots in each of the 357 qubit-wise-commuting bases that (m_0, m_1, m_2) require (1.8×10^7 shots per repetition). Over 8 repetitions the mean estimates $(\widehat{m}_0, \widehat{m}_1, \widehat{m}_2) = (2.730, 10.55, 47.01)$ lie 0.87, 0.72 and 0.66 standard errors (spread over repetitions divided by $\sqrt{8}$) from the exact $(2.737, 10.61, 47.64)$; for the $L = 4$ chain (8 qubits, 126 bases) every moment is recovered within 0.06 standard errors. The number of bases is set by the Pauli support of O_k , not by the moment values, so it can be counted without the statevector once the probe shift $\langle \rho_\pi \rangle$ is known. That shift vanishes by the reflection symmetry of the open chain: for even L , reflection maps ρ_π to $-\rho_\pi$, so $\langle \rho_\pi \rangle = 0$ in a reflection-symmetric ground state. The ground states are not half filled ($N = 2, 4$ and 5 electrons at $L = 4, 6$ and 8, where $\langle \rho_\pi \rangle = 0$ was checked numerically; it is assumed at $L = 10$ and 12). The committed script `estimator_scaling.py` counts the bases this way and reproduces the count from the full eigendecomposition where affordable ($L \leq 8$). The full (m_0, m_1, m_2) battery needs 126, 357, 721, 1191, 1837 bases at $L = 4, 6, 8, 10, 12$ (8 to 24 qubits); the (m_0, m_1) pair alone needs 19, 35, 51, 67, 83, growing roughly linearly. The number

of settings thus grows polynomially in this range, not exponentially. We do not turn this into a deployability claim: the counts-based recovery was validated only up to 12 qubits and the basis count against full diagonalization only up to $L = 8$; the shot budget and accuracy at 24 qubits remain to be characterized; and a device would have to prepare the ground state that the classical check takes as given.

The estimator, including its off-diagonal content, is therefore measurable from a fixed shot budget in principle, but not on present hardware. Under depolarizing noise the estimate of m_1 is pulled upward, toward $\text{Tr}(O_1)/\dim \mathcal{H} = 34.3$ against $m_1 = 10.6$ for this probe. Under global depolarizing noise of strength p , the estimate of $m_1 = \langle 0|O|0 \rangle$, with $O = c_{0\uparrow}(H - E_0)c_{0\uparrow}^\dagger$, becomes $(1 - p)m_1 + p \text{Tr}(O)/\dim \mathcal{H}$, a bias of $p[\text{Tr}(O)/\dim \mathcal{H} - m_1]$. For the $L = 8$ device Hamiltonian $\text{Tr}(O)/\dim \mathcal{H} = 6.80$ against $m_1 = 2.10$ (ratio 3.24), so the bias is upward. For the density probe, an exact density-matrix simulation with local two-qubit depolarizing noise at $\varepsilon = 2.5 \times 10^{-3}$ on 292 CZ gates puts the bias at 37% of m_1 ; this is exact for that surrogate and a lower bound for a real circuit. A coherent-ZZ error model gives 47% and global depolarizing 116% (Sec. 6). No off-diagonal moment has been measured on a device in this work. A job script that builds the rotated-basis circuits and applies this estimator (`hardware_blind_job.py`) is in the repository; its state preparation is a placeholder, it has not been run on a quantum processor, and an off-diagonal test on hardware awaits lower gate error.

We close with a Monte-Carlo calibration of this test (Fig. 14) at $N_s = 5 \times 10^4$, the largest per-circuit shot budget of the `ibm_fez` jobs. It is a simulation: $M = 4 \times 10^3$ replicas of N_s shots are drawn from the exact ground-state distribution $|\langle x|0 \rangle|^2$ of the doped $L = 6$, $U/t = 4$ ring, a system not run on hardware; $\widehat{m}_1$ is the local estimator above for its current probe, and each truncated reconstruction is built from the ground state of the truncated Hamiltonian. The Monte Carlo checks the analytic interval rather than replacing it. The realized estimator is close to Gaussian (skewness -0.04 , excess kurtosis -0.05), its spread matches the analytic shot term to 1% (0.0799 against 0.0806), and its false-positive rate, 4.5% (Wilson 95% interval $[3.9, 5.2]\%$), is consistent with the nominal 5% (Fig. 14a). Two inputs are modeled: the shot term uses the exact-state local-estimator variance, which assumes known amplitudes and omits ratio-estimator and Pauli-grouping overhead, so it likely understates the per-shot uncertainty of a device estimator; the bias term b_k is assumed and swept, not measured. For this probe $\text{Var}(O_1) \approx 3.2 \times 10^2$ gives a shot standard error $\sqrt{\text{Var}(O_1)/N_s} \approx 0.081$ (1.9% of m_1); with the 2% bias budget $\tau_1 = 0.229$ (5.4% of m_1), close to the 5% relative line of Figs. 7 and 8.

Power is high but not uniform in the truncation depth. At the shot-only threshold m_1 alone reaches power ≥ 0.95 at 161 of the 185 truncations, already at $d = 181$ (97% coverage, power 0.98). The exceptions are the four mildest truncations ($d = 182$ –185, one to four determinants dropped) and three narrow windows, $d = 84$ –88, 93–98 and 158–166 (coverage 45–47%, 50–53%, 85–89%), at whose centres power falls to the false-positive level (5.5%, 6.1%, 5.0% at $d = 86, 95, 162$). The 22-point grid of Fig. 14b touches the first two windows only at their edges ($d = 88, 98$; power 0.93, 0.94) and misses the third. An earlier grid-only analysis, with ties broken by floating-point noise, showed a single low-power point near 53% coverage; the scan at every d supersedes it. The noise-free residuals agree: at the deployed 2% bias budget $|\widehat{m}_1 - \bar{m}_1| < \tau_1$ for 17 of the 185 truncations ($d = 183$ –185, 159–165, 94–97, 85–87), and at a 4% budget for 27.

Where the cut falls matters. The probabilities $|\psi_0|^2$ of this sector come in exactly degenerate shells of 6, 12 or 24 determinants, and which members of a shell a truncation keeps is a convention. All the window cuts, and 19 of the 21 truncated grid points, fall inside a shell. Our convention is deterministic (descending $|\psi_0|^2$, ties broken by sector index), but verdicts near the windows depend on it. At $d = 98$ it gives $|\widehat{m}_1 - \bar{m}_1| = 0.284 > \tau_1$, while 300 random choices of the 8 kept members of the 24-fold shell give 0.172–0.295 and rejection by m_1 alone

in 57% of cases; an earlier run with ties broken by floating-point noise found 0.225, a miss. At the ten cuts that fall on a shell boundary ($d = 6, 18, 30, 42, 66, 90, 114, 138, 150, 174$), and so do not depend on the convention, the noise-free residual of m_1 alone exceeds τ_1 at the 2% budget.

Detection sets in at much higher coverage here than in the independence figures, whose idealized full-state and shared-state curves cross their 5% line near 7% and 18% retained determinants ($d \approx 1.6\text{--}4.4 \times 10^4$ of 2.4×10^5 ; Figs. 7 and 8). This reflects a different test, not a sharper one: those curves score the large $L = 12$ sector by a *relative* discrepancy $|\widehat{m}_1 - \bar{m}_1|/\widehat{m}_1$ against a fixed line, this Monte Carlo the small $L = 6$ sector by an *absolute* threshold set by the shot budget. The coverage percentages thus come from different rulers on different-sized sectors (and, at $L = 12$, two reference states), not from one test giving several answers.

The direction of the residual uncertainty matters for a test that can only reject. Underestimating δ_k shrinks τ_k : safe against missed errors (false negatives, acceptable for a necessary condition) but anti-conservative against false rejections (false positives, the costly error for such a test). The false-positive rate above relies on $\bar{m}_1$ being exact at full support ($|\widehat{m}_1 - \bar{m}_1| \sim 10^{-15}$), so that the null hypothesis holds exactly there. For m_2 it does not: the 186-determinant support contains all of φ but not $H\varphi$, so the reconstruction on it is wrong at second order, $\Delta_2 = 0.62$ ($0.48 \tau_2$ at the shot-only threshold, $0.34 \tau_2$ at the deployed one), the boundary-leakage term above. An m_2 rejection of it would be a correct detection, not a false positive; the Monte Carlo tests m_1 only. Away from full coverage the threshold should be set on $\text{Var}(\widehat{m}_k - \bar{m}_k)$, for which $\text{Var}(O_k)$ is a proxy valid only when the reference is trusted (same-sample statistics above; Sec. 9).

The second moment catches some lone- m_1 misses, not all. In the noise-free scan at the 2% budget, m_2 rejects 3 of the 17 truncations that m_1 alone passes ($d = 97, 96, 85$). The clearest is $d = 97$ (52% coverage): in the deterministic order $|\widehat{m}_1 - \bar{m}_1| = 0.186 < \tau_1 = 0.229$ but $|\widehat{m}_2 - \bar{m}_2| = 2.73 > \tau_2 = 1.85$, whatever the tie convention (m_1 alone rejects in none and the battery in all of 300 random shell choices). The full (m_0, m_1, m_2) battery nonetheless passes 12 truncated reconstructions at the 2% budget: $d = 183\text{--}185$ (98–99.5% coverage), 160–165 (86–89%), and 86, 87, 95 (46–51%). Some of these passes are tie-convention effects (the battery rejects in 44%, 40% and 11% of random shell choices at $d = 87, 160$ and 95), but not those at $d = 86$ (1.3%), 161–165 (at most 0.3%) or 183–185 (none). Without a bias budget the battery catches the mid-coverage windows but still passes $d = 161\text{--}164, 184$ and 185; with a 4% budget it passes 22 truncations. The joint battery is thus a partial remedy for the single-moment windows: with finite thresholds no battery of necessary conditions is free of blind spots, and at the deployed order this one has blind spots that we can list.

Two further boundaries frame the demonstration. First, the discrepancy is measured against the exact moments, which serve as an oracle: the demonstration calibrates the interval where classical ground truth exists, and a device-side version that propagates measured rather than assumed bias remains to be done. Second, moment-matrix certificates [81, 82] bound ground-state properties, whereas here two estimates of the same moments are compared and a discrepancy beyond threshold rejects; the screen never certifies. The sum rules are textbook [35, 74] and sum-rule residuals already serve as validation metrics for quantum-simulated spectra [40]; this paper adds their use as reconstruction-independent, interval-valued rejection criteria for sample-based spectral functions, with thresholds fixed in advance.

C The Christoffel miss-distance: proof and evaluation

We re-derive the equality of the extremal cumulative-weight spread with the Christoffel function from three classical lemmas [66, 67, 69]; nothing in this derivation is new. Write

$K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ for the Christoffel–Darboux kernel and $\lambda_n(t) = 1/K_n(t, t)$ for the Christoffel function, and assume the moment vector is interior to the moment cone, $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$ (for a finite N -atom Lehmann spectrum this holds for $n \leq N-1$). The measures ν compared below are positive measures on $\mathbb{R}$. Restricting them to the Lehmann half-line $\omega \geq 0$ can only narrow the bracket, so W_n remains an upper bound there.

Lemma 1 (variational identity). $\lambda_n(t) = \min\{\int |P|^2 d\mu : P \in \mathcal{P}_n, P(t) = 1\}$, with minimizer $P^* = K_n(\cdot, t)/K_n(t, t)$. Expanding $P = \sum_{j=0}^n c_j p_j$ gives $\int |P|^2 d\mu = \sum_j c_j^2$ while $1 = P(t)^2 = (\sum_j c_j p_j(t))^2 \leq (\sum_j c_j^2) K_n(t, t)$ by Cauchy–Schwarz, with equality at $c_j = p_j(t)/K_n(t, t)$; the minimum is $1/K_n(t, t) = \lambda_n(t)$.

Lemma 2 (maximal atom). Among positive measures ν with $\int \omega^k d\nu = m_k$ for $k \leq 2n$, the supremum of the mass at a point t is $\sup_\nu \nu(\{t\}) = \lambda_n(t)$. For the extremal P^* of Lemma 1, $\nu(\{t\}) = P^*(t)^2 \nu(\{t\}) \leq \int (P^*)^2 d\nu = \int (P^*)^2 d\mu = \lambda_n(t)$, the middle equality using $\deg(P^*)^2 = 2n$ and that ν shares those moments. If $p_n(t) \neq 0$ the bound is attained by the $(n+1)$ -point Gauss–Radau rule with a node prescribed at t : a positive measure matching $m_0, \dots, m_{2n}$ whose weight at t is $\lambda_n(t)$. At the n zeros of p_n this rule does not exist (one node escapes to infinity), and the supremum is approached but not attained.

Lemma 3 (ordinate gap). Let $p_n(t) \neq 0$, and let $x_0 < \dots < x_n$ be the nodes of that Gauss–Radau rule, one of them equal to t , with weights $\lambda_n(x_i) > 0$. For every positive ν sharing $m_0, \dots, m_{2n}$, with $F_\nu(t) = \nu((-\infty, t])$, the Chebyshev–Markov–Stieltjes inequalities give

$$\sum_{x_i < t} \lambda_n(x_i) \leq F_\nu(t^-) \leq F_\nu(t) \leq \sum_{x_i \leq t} \lambda_n(x_i)$$

[66, 67, 100, 101]. They follow from polynomials $Q_\pm \in \mathcal{P}_{2n}$ with $Q_- \leq \mathbf{1}_{(-\infty, t)}$ and $\mathbf{1}_{(-\infty, t]} \leq Q_+$ that coincide with these indicators at the nodes, so that $\int Q_\pm d\nu = \int Q_\pm d\mu$ equal the two sums. The bracket has width $\lambda_n(t)$. Its upper end is attained by the Radau rule itself; its lower end is approached by Radau rules whose prescribed node t' decreases to t . Hence $\sup_{\nu_1, \nu_2} |F_{\nu_1}(t) - F_{\nu_2}(t)| = \lambda_n(t)$. At the zeros of p_n the same supremum follows from the continuity of λ_n and the right-continuity of F .

Combining the three, the extremal spread equals $\lambda_n(t) = W_n(t)$ for every real t , and any single reconstruction consistent with the moments obeys $|F_{\text{rec}}(t) - F_{\text{true}}(t)| \leq W_n(t)$. No convex-hull hypothesis is needed: the Radau rule has positive weights for every t with $p_n(t) \neq 0$, inside or outside the convex hull of the support. We checked this on the $U/t = 8$ current measure of Fig. 13: 24 Radau rules ($n = 1, \dots, 4$ at six values of t , 20 of them outside the convex hull of the weighted support) all have positive weights, place mass $W_n(t)$ at t to a relative error of 2×10^{-16} , and match $m_0, \dots, m_{2n}$ to 1.4×10^{-17} . Monotonicity $W_{n+1} \leq W_n$ follows from $K_{n+1} = K_n + p_{n+1}^2 \geq K_n$.

Pointwise vanishing $W_n(t) \rightarrow 0$ is a thermodynamic-limit statement. If the discrete spectrum densifies to a measure that is Stahl–Totik regular with compact support S and absolutely continuous with a density A obeying the local Szegő condition $\int_I \log A > -\infty$ on an interval $I \subset S$, then $n W_n(x) \rightarrow A(x)/\omega_S(x)$ for almost every $x \in I$, with ω_S the equilibrium density of S [83]; on a single interval this reduces to Máté–Nevai–Totik [97].

Evaluation of the bound from the moments. The Christoffel function is computed from the moments $\{m_k\}_{k=0}^{2n}$ by a short pipeline [69, 71]. With the inner product $\langle f, g \rangle_\mu = \int f(\omega)g(\omega)d\mu(\omega)$, so that $\langle \omega^a, \omega^b \rangle_\mu = m_{a+b}$, Gram–Schmidt on $\{1, \omega, \dots, \omega^n\}$ gives the orthonormal polynomials, with $p_0 = 1/\sqrt{m_0}$; they obey the three-term recurrence

$$\beta_{j+1} p_{j+1}(\omega) = (\omega - \alpha_j) p_j(\omega) - \beta_j p_{j-1}(\omega), \quad (\text{C.1})$$

with $\alpha_j = \langle \omega p_j, p_j \rangle_\mu$ and $\beta_j = \langle \omega p_j, p_{j-1} \rangle_\mu$, all fixed by $\{m_k\}$. The map from power moments to (α_j, β_j) becomes ill-conditioned as n grows, so high orders need high-precision arithmetic.

The nodes and weights of the $(n+1)$ -point Gauss rule are the eigenpairs of the Jacobi matrix $\mathbf{J}_{n+1} = \text{tridiag}(\beta_j, \alpha_j, \beta_j)$ (Golub–Welsch), and $K_n(t, t) = \sum_{j=0}^n p_j(t)^2$, evaluated by the same recurrence, gives $W_n(t) = 1/K_n(t, t)$. Every quantity in the bound is thus an explicit function of the moments, readable at any frequency.

We evaluate it on the $L = 6$, $U/t = 8$ current measure of Fig. 13, normalized to $m_0 = 1$; five of its atoms carry weight. (Its ground level is a spin triplet; the current measure is the same for all three states, because J is a spin scalar.) The numbers below come from `src/export_christoffel_dat.py` and `src/christoffel_tolerance_lp.py` (data/2026-09-27_theory_numerics.json). We run the recurrence in 60-digit arithmetic on the atoms and weights of the measure, which fix the same coefficients as its moments, and locate the maxima by root-finding: $\max_t W_n = 1, 0.969, 0.886, 0.882$ for $n = 1, \dots, 4$. The $n = 1$ maximum equals m_0 and lies at the mean frequency $\bar{\omega} = 11.10 t$, the zero of p_1 , so at the deployed order (m_0, m_1, m_2) the bracket is vacuous at the spectral peak. At the dominant atom ($\omega = 11.09 t$, residue 0.778), W_n reaches the residue at $n = 4 = N - 1$, and it cannot go lower at any order, because Lemma 2 with $\nu = \mu$ gives $W_n(t) \geq \mu(\{t\})$: no number of moments narrows the bracket at the peak below 0.778.

The figure's abscissa $t \in [-1, 1]$ is the affine image of the frequency range $\omega \in [0.148, 52.21] t$, the positive excitation energies of the full Fock space, zero-weight states included. The weighted support $[2.49, 13.92] t$ maps to $[-0.910, -0.471]$, and 73.5% of $[-1, 1]$ lies above the last weighted pole. The affine map moves points along the axis but does not change W_n , which is a mass. The rescaled points 0.3, 0.6, 0.9 correspond to $\omega = 34.0, 41.8, 49.6 t$, far above the band. There $W_1 = 4.0 \times 10^{-3}, 2.2 \times 10^{-3}, 1.4 \times 10^{-3}$, and at 0.6 the higher orders give $W_2 = 1.4 \times 10^{-5}, W_3 = 2.7 \times 10^{-8}, W_4 = 3.6 \times 10^{-10}$. For $n = 1$ these small values are Cantelli's one-sided Chebyshev bound, $W_1(\omega) = m_0 \sigma^2 / [\sigma^2 + (\omega - \bar{\omega})^2]$ with σ^2 the variance of the normalized measure (it agrees with the orthogonal-polynomial evaluation to 1.6×10^{-15}). They reflect the distance from the mean, not a feature of this spectrum. Just above the top pole, at $\omega = 14 t$, $W_1 = 0.20$.

All of this assumes exact moments. If the moments are matched only within the deployed tolerances, $|\Delta_k| \leq \tau_k$ for $k \leq 2$, the largest atom a positive measure can place at ω is a linear program. With thresholds from the deployed rule applied to this state ($\tau_0, \tau_1, \tau_2 = 0.051, 0.57, 6.5$ before normalization), it is $3.2 W_1$ (0.63 of m_0) at $\omega = 14 t$, $7.7 W_1$ at $20 t$, and 10.1 – $10.7 W_1$ at the rescaled points 0.3–0.9. These τ_k use the lower-bound shot variance of App. B; wider, realistic tolerances would inflate the atom further. At the deployed order a passing reconstruction is therefore bracketed much more loosely than W_1 alone suggests.

Evaluation of the Markov–Krein window bound (Proposition 1). On the same $L = 6$, $U/t = 8$ current measure (committed script `markov_krein_window.py`), at exact moments: at $n = 4$ ($m_0, \dots, m_8$), a window with one edge at the 400-point-grid maximizer of W_4 , just above the dominant atom, gives $\mu \in [0.070, 0.945]$, width 0.875; the exact maximum of W_4 , at a nearby frequency, is 0.882. A window that contains the peak with both edges in weight-free frequency (rescaled $[-0.75, -0.35]$) gives $\mu \in [0.9732, 0.9758]$ around the exact 0.9744, and the empty window rescaled $[0.3, 0.9]$ ($\omega \approx 34$ – $50 t$) gives $\mu \in [0, 3.2 \times 10^{-9}]$. At the deployed order $n = 1$ (m_0, m_1, m_2) the peak-containing window gives $\mu \in [0.848, 1]$ and the empty window $\mu \in [0, 0.004]$, while a window with an edge on the peak is uninformative (its bracket runs from 0 to within 2×10^{-4} of m_0). These are exact-moment brackets. Under the deployed tolerances the upper end for the empty window is at least 0.040, the largest atom that a measure with m_0, m_1, m_2 within tolerance can place at its lower edge ($\omega \approx 34 t$).

D Complementary state-level check: the Gauss law

The Gauss-law check mentioned in Results I is not a moment test; we record its construction and scope here. In the U(1) lattice Schwinger model the generators G_x of gauge transformations commute with the Hamiltonian, $[H, G_x] = 0$, and every physical state satisfies $G_x|\psi\rangle = 0$ at every site, so $\langle \sum_x G_x^2 \rangle = 0$ exactly. The operator $\sum_x G_x^2$ is diagonal in the computational basis, so on hardware it could be estimated from computational-basis samples of the prepared state, with no reconstruction and no moment. It tests the prepared state, not a reconstructed spectrum: it responds to leakage out of the physical sector, an error class the moment test does not target.

Figure 15 illustrates the identity on a small instance: the quantum-link form of the model with $N = 4$ staggered matter sites and 3 spin- $\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$ ($[H, G_x] = 0$ to machine precision). The plotted state is $|\psi(\epsilon)\rangle \propto (1 - \epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ an unphysical state. The parameter ϵ is an amplitude, not a weight: the unphysical weight is $f_u(\epsilon) = \epsilon^2 / [(1 - \epsilon)^2 + \epsilon^2]$, 0.059 at $\epsilon = 0.2$ and 0.155 at $\epsilon = 0.3$. Because $\sum_x G_x^2|\psi_0\rangle = 0$ and $\langle\psi_0|\phi_u\rangle = 0$, the violation is $\langle \sum_x G_x^2 \rangle = g_u f_u(\epsilon)$ with $g_u = \langle\phi_u|\sum_x G_x^2|\phi_u\rangle$. The curve is therefore fixed by construction: zero for the physical state, rising monotonically with ϵ , with its scale set by g_u alone. It illustrates the identity and does not measure detection power. The values are computed exactly on the state vector; no sampling is simulated.

The admixed state is chosen by an explicit rule. The generator `src/export_gauss_state.py` takes $|\phi_u\rangle$ as the lowest-energy eigenstate of H among charge-neutral states with the smallest nonzero violation, $\sum_x G_x^2 = 2$ (one site with $G = +1$ and one with $G = -1$; a 27-dimensional subspace). Every state in that subspace has $g_u = 2$, so the plotted curve is $2f_u(\epsilon)$. The figure was first made with a different rule: the unphysical eigenstate of H with the largest overlap $|\langle k|M|\psi_0\rangle|$, where $M = \sum_n (-1)^n n_n$ is the staggered occupation. That rule is ill-posed. M is gauge-invariant, so every such overlap vanishes (the largest is 1.8×10^{-15}), and the pick, set by rounding noise, happened to be a state with $g_u = 2$. The explicit rule reproduces the plotted values to all five printed decimals (`data/2026-09-27_gauss_state.json`). Over the 124 unphysical eigenstates of this model g_u ranges from about 1 to 10, so the slope of the curve reflects this choice. An earlier sweep under an independent bit-flip channel on every qubit (`data/2026-08-18_gausslaw_falsifier.json`) also gives $\langle \sum_x G_x^2 \rangle > 0$ at every nonzero flip rate (0.25 at $p = 0.05$). It averages only 40 sampled flip patterns per rate and is not monotone in p at that sample size (3.05 at $p = 0.35$, 2.52 at $p = 0.40$), so we do not use it as a graded curve.

Gauss-law violation has been measured in a 71-site cold-atom quantum simulator [39] and analyzed theoretically under gauge-breaking errors [125]. The lattice Schwinger model is also the model of self-verifying variational simulation on a trapped-ion analog quantum simulator [41] and of early digital simulations of Schwinger-model dynamics [126, 127].

E Within-sector redistributions: injectivity and the reach of the deployed battery

This appendix proves Theorem 2 and evaluates it on one spectrum. The independent estimator detects weight that leaks *outside* the retained determinant support, but not a redistribution strictly *within* it that preserves the moments used (Sec. 4). We treat that residual in three steps: *injectivity* (part (a): in exact arithmetic, $m_0, \dots, m_{N-1}$ fix the weights on N distinct poles); *conditioning* (part (b) and the Corollary: a sufficient and a necessary condition for exposing a redistribution at finite shots); and *reach* (a linear program and explicit counterexamples on

the doped $L = 6$, $U/t = 4$ current spectrum, with thresholds from App. B; Table 3). The result at the deployed order is negative: the battery (m_0, m_1, m_2) has no guaranteed power against within-sector redistributions of this spectrum, although its poles are well separated. We characterize this residual; we do not close it. Unless stated otherwise, the numbers below are in data/2026-09-27_theory_numerics.json (script src/within_sector_lp.py).

Injectivity (proof of Theorem 2(a)). A $T = 0$ Lehmann current spectrum on a finite sector is a discrete measure $A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n)$ on N distinct poles $\{\omega_n\}$, where $w_n = \sum_{a: E_a - E_0 = \omega_n} |\langle a | J | 0 \rangle|^2 \geq 0$ is the total weight of the eigenstates at ω_n . Its moments are $m_k = \sum_n w_n \omega_n^k$, i.e. $\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$ a Vandermonde matrix. Because the poles are distinct, the block $V_{0:N-1}$ of rows $0, \dots, N-1$ is invertible, so the weight-to-moment map is injective through order $N-1$: a redistribution $\mathbf{w} \rightarrow \mathbf{w} + \delta$ with $\delta \neq 0$ cannot preserve all of $m_0, \dots, m_{N-1}$. A redistribution that preserves $m_0, \dots, m_K$ ($K < N-1$) zeroes the first $K+1$ entries of $V_{0:N-1}\delta$, so $\|(\Delta_{K+1}, \dots, \Delta_{N-1})\|_2 = \|V_{0:N-1}\delta\|_2 \geq \sigma_{\min}(V_{0:N-1})\|\delta\|_2$; the largest of these $N-1-K$ shifts is at least this norm divided by $\sqrt{N-1-K}$, which is Eq. (2). If the poles may also move, an N -atom measure has $2N$ parameters and is fixed by $m_0, \dots, m_{2N-1}$ (Prony) [65, 102]. Weight traded among degenerate eigenstates at a single frequency changes no moment and leaves $A(\omega)$ unchanged; it is not an error.

For the $L = 6$, $U/t = 4$ spectrum the sixteen eigenstates with nonzero weight fall on $N = 5$ distinct frequencies, $\omega_n/t = 2.73, 5.57, 7.83, 8.70, 10.66$, with weights 0.040, 0.004, 0.393, 0.116, 0.001 ($m_0 = 0.554$; the pole at $7.83t$ carries 71% of it). The map is therefore injective through m_4 : a redistribution among these poles that preserves $m_0, \dots, m_K$ changes some m_j with $K < j \leq 4$ (at most m_9 if the poles also move). In exact arithmetic the residual is thus resolved by m_4 . That is two orders beyond the deployed battery, and it says nothing about detection at finite shots. Equation (2) is valid here but loose: for the redistribution CE1 below, its right-hand side is 3.8×10^{-3} , while the larger of the shifts in m_3 and m_4 is 85. The sufficient condition of Theorem 2(b)(i) is far from met here: for CE1 and CE2 the right-hand side of (2) is below 4×10^{-3} , against $\max_j \tau_j = \tau_4 = 122$ in the same units.

Conditioning, and when detection is possible (proof of Theorem 2(b)). Invertibility of $V_{0:N-1}$ holds in exact arithmetic. At finite shots a shift counts only if it exceeds the threshold τ_j . Two bounds, in opposite directions, turn detectability into a statement about the size of δ and the geometry of the poles it touches.

Sufficient condition. By Eq. (2), a redistribution preserving $m_0, \dots, m_K$ is exposed (its noise-free shift exceeds the threshold) at some order $j \leq N-1$ if $\sigma_{\min}(V_{0:N-1})\|\delta\|_2/\sqrt{N-1-K} > \max_{K < j \leq N-1} \tau_j$. Since $\|\delta\|_1/\sqrt{N} \leq \|\delta\|_2 \leq \|\delta\|_1$, this is a condition on $\rho = \|\delta\|_1$ up to a factor $\sqrt{N}$. It is informative only when $\sigma_{\min}$ is not small, and it uses moments up to order $N-1$.

Necessary condition. Let δ preserve $m_0, \dots, m_{s-2}$ and be supported on poles inside an interval of diameter D and midpoint ω_c . (Among s poles, preserving $m_0, \dots, m_{s-2}$ leaves a one-parameter family, and m_{s-1} is the first moment it changes.) Since $\sum_n \delta_n q(\omega_n) = 0$ for every polynomial q of degree at most $s-2$, the signed shift is $\sum_n \delta_n \omega_n^{s-1} = \sum_n \delta_n (\omega_n - \omega_c)^{s-1}$, hence

$$|\Delta_{s-1}| \leq \rho (D/2)^{s-1}.$$

Subtracting from ω^j its Taylor polynomial of degree $s-2$ at ω_c gives, for every $j \geq s-1$, $|\Delta_j| \leq \rho \binom{j}{s-1} (|\omega_c| + D/2)^{j-s+1} (D/2)^{s-1}$ (Lagrange remainder); expanding instead $\omega^j = \sum_i \binom{j}{i} \omega_c^{j-i} (\omega - \omega_c)^i$, whose terms with $i \leq s-2$ are combinations of preserved moments and vanish, and bounding the rest by $|\omega_n - \omega_c| \leq D/2$ gives the term-by-term form of Theorem 2(b)(ii). Either way every shift vanishes as D^{s-1} . Exposure at order $s-1$ therefore requires $\rho (D/2)^{s-1} > \tau_{s-1}$. With $\tau = z_{1-\alpha/2} \delta$ and $\delta^2 = \text{Var}(O_{s-1})/N_s + b_{s-1}^2$ this reads

$$N_s > \frac{z_{1-\alpha/2}^2 \text{Var}(O_{s-1})}{\rho^2 (D/2)^{2(s-1)} - z_{1-\alpha/2}^2 b_{s-1}^2},$$

and no budget suffices once $\rho(D/2)^{s-1} \leq z_{1-\alpha/2} b_{s-1}$. The shot cost of detecting a redistribution of fixed size inside a collapsing cluster thus grows at least as $D^{-2(s-1)}$. Numerically, the identity holds to a relative error of 1.5×10^{-12} , and the bound holds in 11 structured cases (CE1 and CE2 below, the six optima of Eq. (E.1) and three committed random cases) and in 20 000 random null-space redistributions on random pole subsets (seed 20260927, no violation). The largest ratio $|\Delta_{s-1}|/[\rho(D/2)^{s-1}]$ is 1.000, 0.499 and 0.231 at $s = 2, 3, 4$, so the bound is tight at $s = 2$. If a cluster contracts uniformly, so that its minimal pole separation falls from $\Delta\omega_{\min}$ to $\Delta\omega$, D shrinks by the same factor and the lower bound (3) on the shot cost grows by at least $(\Delta\omega_{\min}/\Delta\omega)^{2(s-1)}$. A finite separation below which no budget suffices exists only with a bias floor, $b_{s-1} > 0$.

Conditioning numbers. The same collapse appears in $\sigma_{\min}$: for s nodes of fixed relative geometry and diameter $D \rightarrow 0$, $\sigma_{\min}$ of the $s \times s$ Vandermonde block scales as D^{s-1} . The sharp super-resolution results for this regime [103–106] are proved for Fourier (unit-circle) nodes, not for the real nodes used here, for which both directions follow from elementary bounds on the $s \times s$ block V of the cluster. Row n of V^{-1} holds the coefficients of the Lagrange polynomial $\prod_{j \neq n} (\omega - \omega_j)/(\omega_n - \omega_j)$, which gives Gautschi's bound $\|V^{-1}\|_{\infty} \leq \max_n \prod_{j \neq n} (1 + |\omega_j|)/|\omega_n - \omega_j|$ [107, 108] and hence $\sigma_{\min}(V) \geq [\sqrt{s} \max_n \prod_{j \neq n} (1 + |\omega_j|)/|\omega_n - \omega_j|]^{-1}$, a lower bound that scales as D^{s-1} at fixed relative geometry. Conversely, the redistribution that spans the null space of rows $0, \dots, s-2$ has $\|V\delta\|_2 = |\Delta_{s-1}| \leq \rho(D/2)^{s-1} \leq \sqrt{s} \|\delta\|_2 (D/2)^{s-1}$, so $\sigma_{\min}(V) \leq \sqrt{s} (D/2)^{s-1}$. A direct check with $s = 2, 3, 4$ equally spaced nodes near $\omega = 7.83 t$ (spacing 0.02 – $0.2 t$) gives fitted exponents 0.995, 1.98 and 2.96 (data/2026-08-24_within_sector_control.json). Condition numbers depend on the frequency unit, so we quote them only on nodes mapped affinely onto $[-1, 1]$ (outermost poles at ± 1): for the five poles above, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$. With a minimal separation of $0.87 t$ these poles are well separated, yet good conditioning does not give the deployed battery guaranteed power: that is set by the dimension of a null space (next paragraph).

Reach of the deployed battery (instantiation of Theorem 2(b) and Corollary). The zero-power clause of part (b) is immediate: a vector in the null space of rows $0, \dots, K+1$ of V , scaled so that $w_n + \delta_n \geq 0$, preserves $m_0, \dots, m_{K+1}$; every smaller rescaling does too, so the smallest shift at order $K+1$ is zero. The deployed battery uses m_0, m_1, m_2 with thresholds $\tau_k = z_{0.975} \delta_k$ from App. B: $\tau_0 = 0.0287$, $\tau_1 = 0.229$, $\tau_2 = 1.848$ at $N_s = 5 \times 10^4$ with a 2% bias budget $b_k = 0.02 m_k$ (τ includes $z_{0.975} = 1.96$, unlike the 1σ half-width δ). The shot term uses the exact-state local-estimator variance, a lower bound on the true per-shot variance, so realistic thresholds would be wider. For a redistribution that preserves $m_0, \dots, m_K$, the largest shift it can impose on the next moment is the linear program

$$\Delta_{K+1}^{\max} = \max_{\delta} \left| \sum_n \delta_n \omega_n^{K+1} \right| \quad \text{s.t.} \quad \sum_n \delta_n \omega_n^j = 0 \quad (j \leq K), \quad \delta_n \geq -w_n, \quad (\text{E.1})$$

the constraint $\delta_n \geq -w_n$ keeping the corrupted weights non-negative; together with $\sum_n \delta_n = 0$ it also bounds $\|\delta\|_1 \leq 2m_0$.

Table 3 gives the full range of the signed shift. A redistribution that preserves m_0, m_1 can move m_2 by up to $7.09 = 3.8 \tau_2$, so some such redistributions are exposed at m_2 . The guaranteed power, however, is set by the smallest shift a redistribution of a given size can produce, and that is zero. Rows $0, 1, 2$ of V have a null space of dimension $N - 3 = 2$: some redistributions preserve m_0, m_1 and m_2 exactly, and scaling one of them down gives an exactly invisible redistribution of every smaller size. Three explicit cases, all positive measures that pass the deployed battery: (i) CE1 preserves m_0, m_1, m_2 exactly and moves 72% of the weight ($\|\delta\|_1/2 = 0.72 m_0$); it empties two of the five poles and lowers the dominant residue from 0.393 to 0.037. (ii) CE2 preserves m_0, m_1 exactly and shifts m_2 by exactly τ_2 ; it moves 86% of

the weight and lowers the dominant residue to 0.073. (iii) CE3 keeps $|\Delta_k| \leq \tau_k$ for $k \leq 2$, has $\|\delta\|_1 = 2.03 m_0$ (above $2m_0$ because m_0 itself moves by τ_0), and empties three poles, the dominant one included. The random $K = 1$ case committed in `src/within_sector_control.py` (seed 0) moves 25% of the weight ($\|\delta\|_1/2 = 0.25 m_0$; its output file lists $\|\delta\|_1/m_0 = 0.49$ as the moved fraction) and shifts m_2 by $0.044 \tau_2$; it passes as well. The guaranteed power of the deployed battery against within-sector redistributions of this spectrum is therefore zero. The reason is the dimension of the null space, not conditioning, and the same holds for any spectrum with four or more distinct poles.

Adding m_3 and m_4 would remove the null space (rows $0, \dots, 4$ have $\dim \ker = 0$), but at this shot budget the extended battery would still miss a redistribution that moves 72% of the weight. Under the same threshold rule $\tau_3 = 14.98$ and $\tau_4 = 122.0$. Among redistributions that preserve m_0, m_1, m_2 exactly, the largest m_3 shift is $5.26 = 0.35 \tau_3$ (Table 3), and CE1, like every scaled-down copy of it, stays below both thresholds ($0.23 \tau_3$ and $0.70 \tau_4$). Some such redistributions would be caught at m_4 : the one that maximizes the m_3 shift moves m_4 by $145 = 1.19 \tau_4$, a marginal case given the lower-bound variance. CE2 and CE3, which use the tolerance on m_2 , would be caught ($|\Delta_3| = 3.3 \tau_3$; $|\Delta_4| = 7.1$ and $7.2 \tau_4$).

The joint battery does catch some errors that m_1 alone misses, but the example is a truncation, not a fixed-pole redistribution: truncating the determinant basis of the same system to the $d = 97$ determinants of largest $|\psi_0|^2$ (52% of the 186-determinant support; App. B) gives a reconstruction that m_1 alone passes and m_2 rejects. The cut falls inside a 24-fold degenerate shell, so the gaps depend on how ties are ordered. The committed smoke test `src/verify.py` (NumPy `argsort` order, which does not fix the order of ties; our machine) gives $|\bar{m}_1 - \widehat{m}_1| = 0.155 < \tau_1$ and $|\bar{m}_2 - \widehat{m}_2| = 2.560 > \tau_2$; the deterministic `lexsort` order of `src/interval_battery.py` gives 0.186 and 2.729; and for all 300 random tie-breaks within the shell m_1 alone rejects none and the battery rejects all (`data/2026-09-27_interval_battery.json`). This is one catch, not a guarantee: in the exhaustive scan of App. B ($d = 1, \dots, 185$, same thresholds, `lexsort` order) the full (m_0, m_1, m_2) -Hankel battery passes 12 truncated reconstructions ($d = 86, 87, 95, 160\text{--}165$ and $183\text{--}185$). Under random tie-breaks within the degenerate shells, ten of these pass in at least 89% of 300 draws, and $d = 87$ and $d = 160$ in 56% and 60%.

Positivity cannot help against within-sector redistributions: a redistribution with non-negative weights is itself a positive measure, so the Hankel and Stieltjes conditions hold by construction. Only independent moments of high enough order, estimated precisely enough, can expose it, and at the deployed order the guaranteed power is zero. We report this residual; we do not claim to close it.

F Device run: reference samplers, line shapes and the n_k channel

This appendix records the reference comparisons (ii) and (iii) and the line-shape comparisons of the $L = 8$ device run (Sec. 6) and the sealed on-device n_k test, which was not run. As in that section, Δ_0 measures support coverage and rewards support spreading; it is not a fidelity diagnostic, and the uniform in-sector sampler, which uses no device, scores lower than the device at every budget.

(ii) *Noiseless sampling at the device's retained counts.* This comparison depends on how the retained counts are allocated. Matched circuit by circuit, the comparator puts 46,216 of its 82,697 shots at 50k on the $t = 0$ circuit, which is a single determinant without noise; its mean Δ_0 is 0.224, 0.279, 0.343 and 0.443 at 50/30/16/4k. With the same total split equally over the seven circuits it is 0.150, 0.205, 0.273 and 0.404. Under both allocations the device lies below every replica at 50, 30 and 16k; at 4k it lies at the 0.6th (per circuit) or 5.8th (equal

split) percentile. The margin shrinks with the budget: the reference mean exceeds the device value 51-, 8.6-, 2.8- and 1.19-fold per circuit, and 34-, 6.3-, 2.2- and 1.08-fold with the equal split.

(iii) *Uniform noise.* Mixing each circuit's output distribution P_k with a fraction p of uniformly random bitstrings, $(1-p)P_k + p/2^{16}$, before post-selection lowers the expected Δ_0 at 50k monotonically, from 0.036 at $p = 0$ to 0.0056 at $p = 0.5$ and 0.0024 at $p = 1$. At 30k and 16k it is lowest near $p = 0.75$ – 0.81 and rises again toward $p = 1$. At 4k it increases with p ($0.270 \rightarrow 0.326$), because post-selection keeps only 6% of uniform shots. At the p that reproduces the device's retention ($p \approx 0.81$), the mixture's mean (0.0023, 0.017, 0.075, 0.295) is below the device at every budget. The device nevertheless lies inside the mixture distribution at 50–16k (85.7th, 91.6th and 88.7th percentiles of 1000 replicas) and at its far upper edge only at 4k (99.9th; one replica lies higher).

Line shape. On the line shape (the relative L_1 error of Sec. 6), too, noiseless sampling does worse at the two higher budgets. Its reconstructions give 0.566 ± 0.032 at 50k (all 80 replicas above the device) and 0.630 ± 0.043 at 30k (device at the 1.25th percentile). At 16k and 4k the device is worse (96th percentile, and above all 80 replicas). The uniform in-sector sampler gives 0.014 at 16k and 0.90 at 4k, below the device at every budget.

A superseded noise model. A simpler model draws shots from $|\varphi|^2$ at the pooled retained count; it is kept in the repository as a superseded record. It underestimates the spread of Δ_0 by factors of 34 to 195 relative to the raw-matched noiseless replicas, and under its own threshold a sampler drawing exactly from $|\varphi|^2$ would fire at every budget (modeled false-positive rate 100%). We use neither that model nor any z -score derived from it. A nonparametric bootstrap of the retained counts is biased upward (its 50k interval $[0.015, 0.101]$ excludes the point estimate 0.0044) and is not used as a test.

The n_k channel. For the momentum distribution n_k of Sec. 6: in the $L = 6$ doped ground state at filling $1/3$ (committed script `nk_falsifier.py`) n_k has a finite-size step $\max_k n_k - \min_k n_k = 0.91$ at $U/t = 4$ and 0.80 at $U/t = 8$; in the thermodynamic limit the one-dimensional step softens to a Luttinger power-law cusp. A reconstruction that flattens the dispersion moves $\bar{n}_k$ toward the filling value, while the true n_k keeps the step. (At half filling the chain is a Mott insulator and n_k is analytic, so this discriminant targets doped sectors, not the half-filled spectrum of Fig. 3.) n_k is a number operator in the momentum basis, read out diagonally after a number-conserving fermionic Fourier transform (a Givens network of $L(L-1)/2$ rotations per spin sector). The particle-number and spin post-selection that protects the coverage channel therefore survives the basis change. The loss of post-selection and the full-space trace-to-value bias that affect the bond-basis rotations do not arise; depolarization acts within the sector instead, as described next. In an in-sector depolarizing model with post-selection at $\varepsilon = 2.5 \times 10^{-3}$, the ~ 60 -CZ transform adds a flattening bias of $\approx 9\%$ of the step (a connectivity-free lower bound, before fermionic-SWAP routing). $\hat{n}_k$ is not immune to noise. Depolarization flattens it toward the same filling value that a dispersion corruption does, so it discriminates only corruptions whose flattening exceeds the device's own, and it inherits the depolarization of the state preparation. Readout error and coherent or leakage noise, which need not flatten toward filling, are not modeled, and post-selection costs shots.

The sealed n_k gate. A pre-registered on-device n_k test was sealed on 2026-09-02 (manifest SHA-256 4962...ae80; doped $L = 6$ ring, $U/t = 8$, 12 qubits), after a simulated Stage-0 gate passed at the manifest's noise floors, $T_1 \geq 150 \mu\text{s}$ and $T_2 \geq 100 \mu\text{s}$ on every qubit of the chain. That pass had little margin: at those floors the sealed post-mitigation residual check ran at 74–100% of its threshold. The single permitted amendment would have relaxed these floors to $T_1 = 100 \mu\text{s}$ and $T_2 = 70 \mu\text{s}$, provided that three pre-committed simulated verdict rows all passed. The worst-admissible row ($T_1 = 100 \mu\text{s}$, $T_2 = 70 \mu\text{s}$, two-qubit error parameter $\varepsilon = 3.5 \times 10^{-3}$) failed on 2026-09-05 at the tightest per-qubit readout cap of a pre-committed

ladder (1.8%): the upper confidence bound of the sealed post-mitigation residual check was 0.0287, against the sealed threshold 0.0267. A row identical except $\varepsilon = 2.85 \times 10^{-3}$ passed (0.0258). In this harness ε is a compound knob that moves the depolarizing and coherent-ZZ components together, so the pair of rows does not locate a physical two-qubit error rate. The amendment was denied, the original floors stand, and the experiment has not been run. The forecasts above and in Sec. 6 assume $\varepsilon = 2.5 \times 10^{-3}$; the failed gate row used 3.5×10^{-3} . An on-device n_k test, like the off-diagonal catch, awaits lower-error hardware.

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Table 2: **What is demonstrated on hardware, in simulation, analytically, and deferred.** The only device-derived quantity is the support set $\mathcal{S}$; no moment is estimated from device counts. Every discriminating result is classically reproducible simulation, and the beyond-classical field test is deferred.

quantity	source	status
support set $\mathcal{S}$ and coverage residual Δ_0 , $L = 8$ addition sector	ibm_fez ($\mathcal{S}$ only)	Δ_0 computed classically on $\mathcal{S}$; rewards support spreading (a uniform in-sector sampler scores lower); tests neither the device nor $A(\omega)$ (Fig. 10)
full-sector recovery, $L = 6$ (companion)	ibm_fez	executability only; the retained determinants all came from device errors (Fig. 11)
off-diagonal moments m_1, m_2 (weight discrimination)	simulation	deferred: forecast depolarizing bias 37–116% of the signal at 292 CZ (Sec. 6; Fig. 9)
momentum distribution n_k (diagonal discriminant)	simulation (sealed gate)	gate-negative at the relaxed noise floors; not run (App. F)
blinded detection and false-positive rates	simulation	sealed primary endpoint: TPR $101/188 = 53.7\%$ at FPR $1/112 = 0.9\%$, unbiased simulated estimator; two sealed secondary predictions failed (Sec. 5)
same-sample $\widehat{m}_k$ and $\bar{m}_k$ (one set of shots)	simulation	correlation $\rho \approx 0.43$ at 84% coverage, no power gain (Sec. 9; App. B)
Christoffel miss-distance bound; moment-interval layer	exact / analytic	classical bound, re-derived; Hausdorff interval shown; composition with the Wang/Mortimer SDP not implemented
field catch of an unknown-truth error (beyond-classical)	—	future work (Sec. 9)

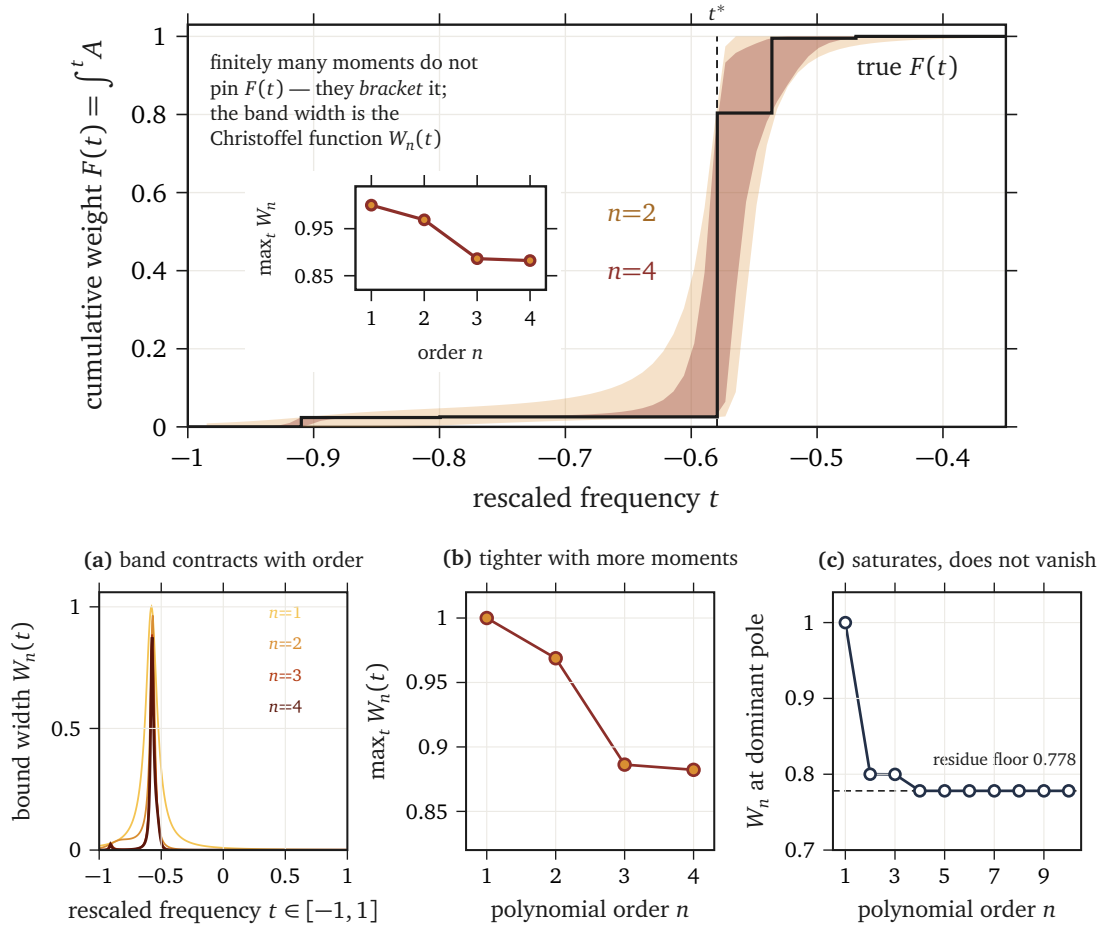


Figure 13: **The Christoffel miss-distance bound at exact moments.** *Top.* The moments $m_0, \dots, m_{2n}$ bracket the cumulative weight $F(t)$ (ink staircase) between extremal Gauss–Radau measures, in a band of Christoffel width $W_n(t)$; the $n=4$ band (deep) nests in the $n=2$ band (light); t^* marks the dominant atom. **(a)** $W_n(t)$ contracts as n grows. **(b)** $\max_t W_n = 1.000, 0.969, 0.886, 0.882$ for $n = 1, \dots, 4$; at the deployed order $n = 1$ the maximum is $W_1 = m_0$, at the mean: the bracket is vacuous. **(c)** At the dominant atom ($\omega = 11.09 t$, residue 0.778) W_n reaches the residue at $n = 4$ and stays there (five weighted atoms). All curves assume exact moments; a reconstruction passing within tolerances is bracketed less tightly. Doped $L = 6$, $U/t = 8$ Hubbard current measure, $m_0 = 1$, rescaled to $t \in [-1, 1]$ (Appendix C) [65, 66, 69, 70, 83, 97].

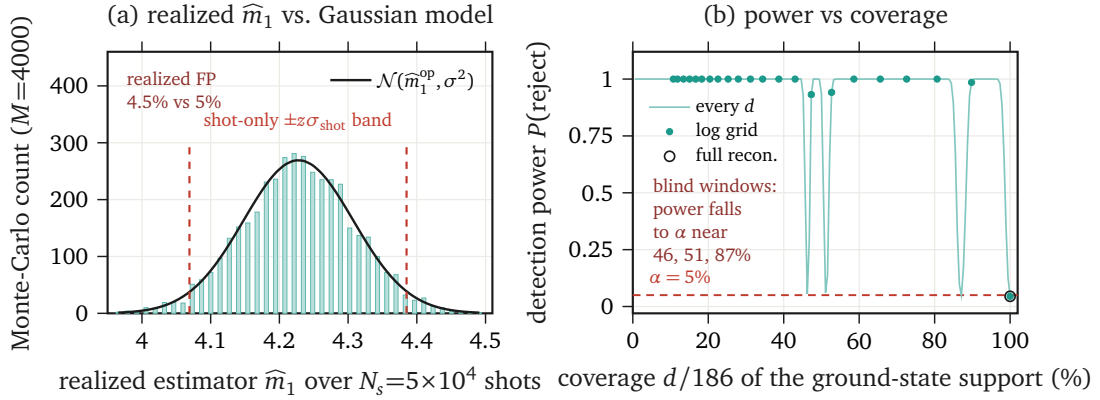


Figure 14: **Monte-Carlo calibration of the interval-moment test** (simulation only, no device data: current probe, doped $L = 6$, $U/t = 4$ ring; $M = 4 \times 10^3$ replicas of $N_s = 5 \times 10^4$ shots from $|\langle x|0\rangle|^2$). **(a)** $\widehat{m}_1$ is near-Gaussian; false-positive rate 4.5% (Wilson 95% interval $[3.9, 5.2]\%$), nominal 5%. Dashed: shot-only band $\pm z\sigma_{\text{shot}} = \pm 0.158$, narrower than the deployed $\tau_1 = 0.229$ (2% bias budget). **(b)** Power of m_1 alone at the shot-only threshold versus retained count d (line; markers: 22-point log grid); coverage $d/186$; open marker ($d = 186$, full support): false-positive rate. Power reaches ≥ 0.95 at 161 of 185 truncations, falling to the false-positive level in windows near 46%, 51% and 87% coverage whose location depends on tie-breaking in degenerate $|\psi_0|^2$ shells (text). The shot term (exact-state local-estimator variance) likely understates a device estimator's per-shot uncertainty.

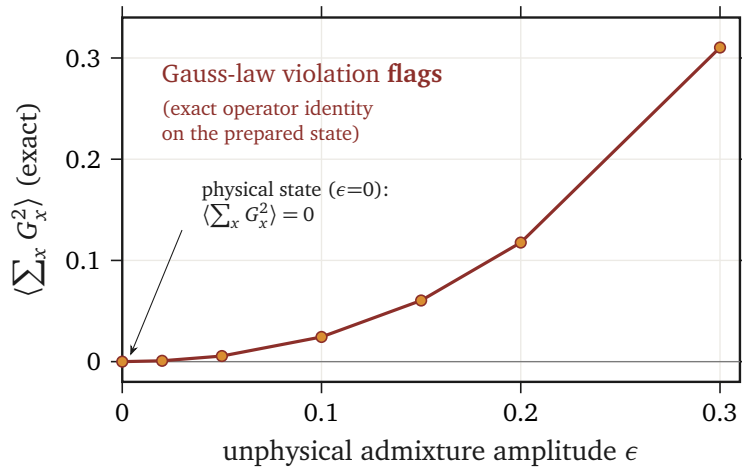


Figure 15: **A state-level check: the Gauss-law identity.** In the U(1) lattice Schwinger model physical states obey $G_x|\psi\rangle = 0$, so $\langle \sum_x G_x^2 \rangle = 0$; $[H, G_x] = 0$ keeps the dynamics in that sector. Plotted: $|\psi(\epsilon)\rangle \propto (1-\epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ the lowest-energy charge-neutral state with the smallest violation, $\sum_x G_x^2 = 2$. The amplitude ϵ gives unphysical weight $f_u = \epsilon^2/[(1-\epsilon)^2 + \epsilon^2]$, only 0.155 at $\epsilon = 0.3$; the curve equals $2f_u$ by construction, computed exactly on the state vector (no sampling is simulated). The check tests the prepared state, not a reconstructed spectrum, and targets gauge-symmetry breaking. $N = 4$ staggered sites, 3 spin- $\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$; generator `src/export_gauss_state.py` (text).

Table 3: Within-sector redistributions of the $L = 6$, $U/t = 4$ current spectrum ($N = 5$ distinct poles). Each row keeps $m_0, \dots, m_K$ exactly and asks how far m_{K+1} can move. “LP range” is the range of the signed shift $\sum_n \delta_n \omega_n^{K+1}$ over all such redistributions with non-negative corrupted weights [Eq. (E.1)]; every range contains 0. Thresholds $\tau_{K+1} = z_{0.975} \delta_{K+1}$ follow the deployed rule ($N_s = 5 \times 10^4$, 2% bias budget, $z_{0.975} = 1.96$; App. B); for m_3 and m_4 , unused in the deployed battery, they are the values the same rule would give. When $\dim \ker V_{0:K+1} > 0$, some finite redistributions leave m_{K+1} exactly unchanged, so the guaranteed power at that order is zero and the maximum describes only the best case.

preserved exactly	next mo- ment	LP range of shift [Eq. (E.1)]	τ_{K+1}	max shift/ τ_{K+1}	$\dim \ker V_{0:K+1}$	in deployed battery?
m_0, m_1	m_2	$[-0.92, 7.09]$	1.848	3.8	2	yes
$m_0, \dots, m_2$	m_3	$[-0.14, 5.26]$	14.98	0.35	1	no
$m_0, \dots, m_3$	m_4	$[-0.30, 0.39]$	122.0	0.003	0	no